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(54) COIL DESIGN FOR INDUCTIVE SENSOR SYSTEM COMPRISING INDUCTIVE TORQUE AND POSITION SENSOR ASSEMBLIES

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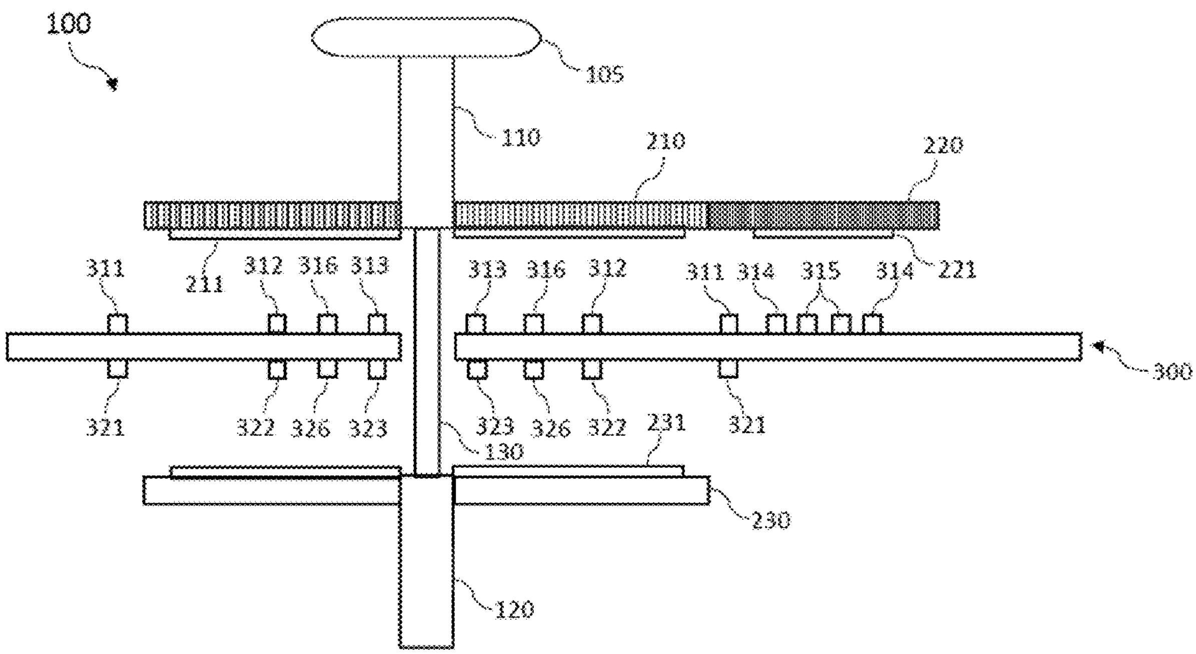
(57) ABSTRACT

An inductive sensor system includes: a circuit board. The circuit board includes: a transmitter coil set including at least a first transmitter coil including a first portion with a first radius and a second portion with a second radius smaller than the first radius and a second transmitter coil including a first portion with a third radius identical to the first radius and a second portion with a fourth radius identical to the second radius. The first transmitter coil and the second transmitter coil are circularly wound in a manner where the first radius and the fourth radius are on a first side of the circuit board while the second radius and the third radius are on a second side of the circuit board. The circuit board further includes: a receiver coil set including at least a first receiver coil and a second receiver coil.



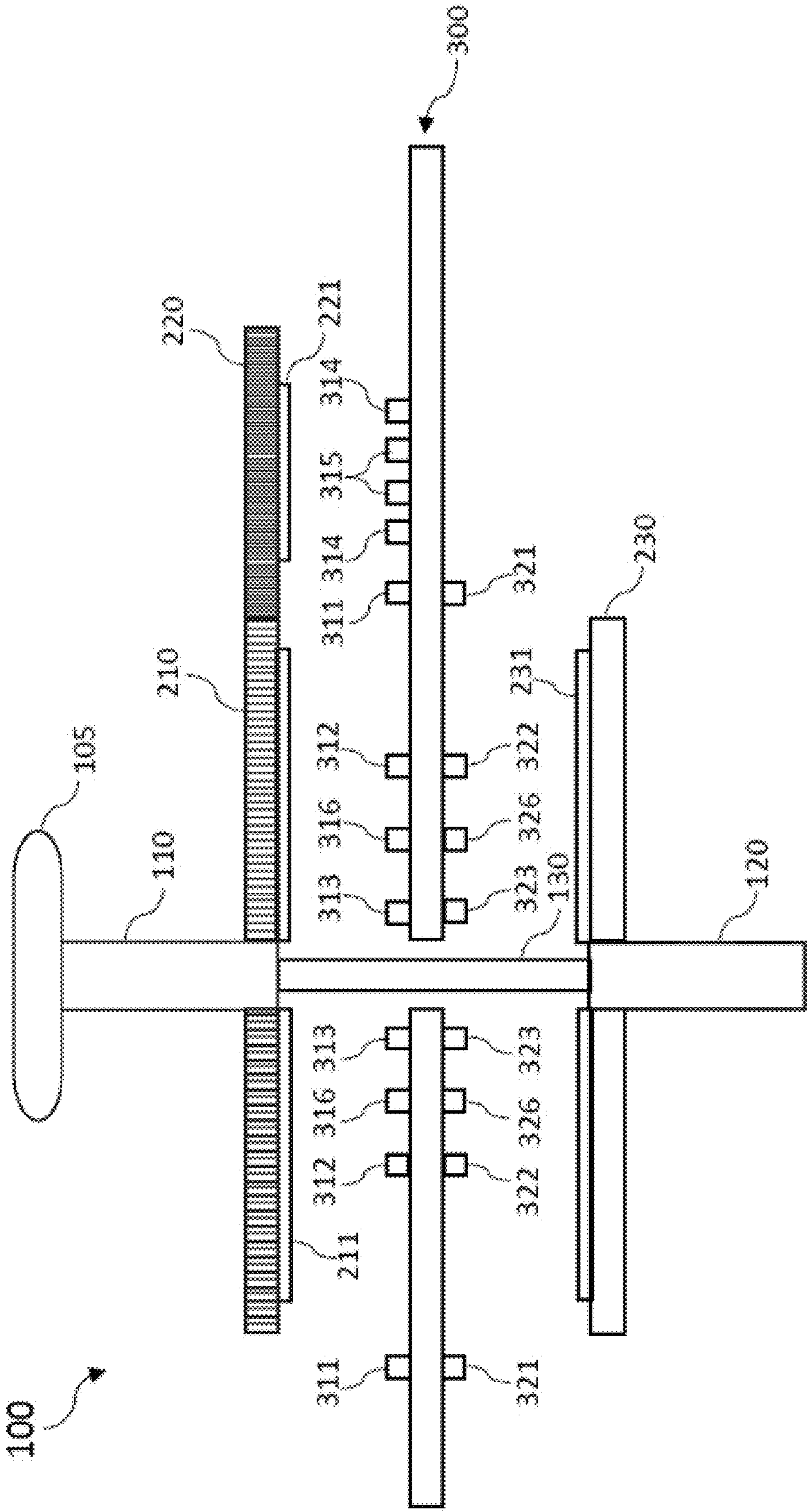

300
220
221
210
105
110
314
315
314
311
321
312
316
313
322
326
323
130
230
231
120
313
316
312
211
323
326
322
311
321
100

FIG. 1

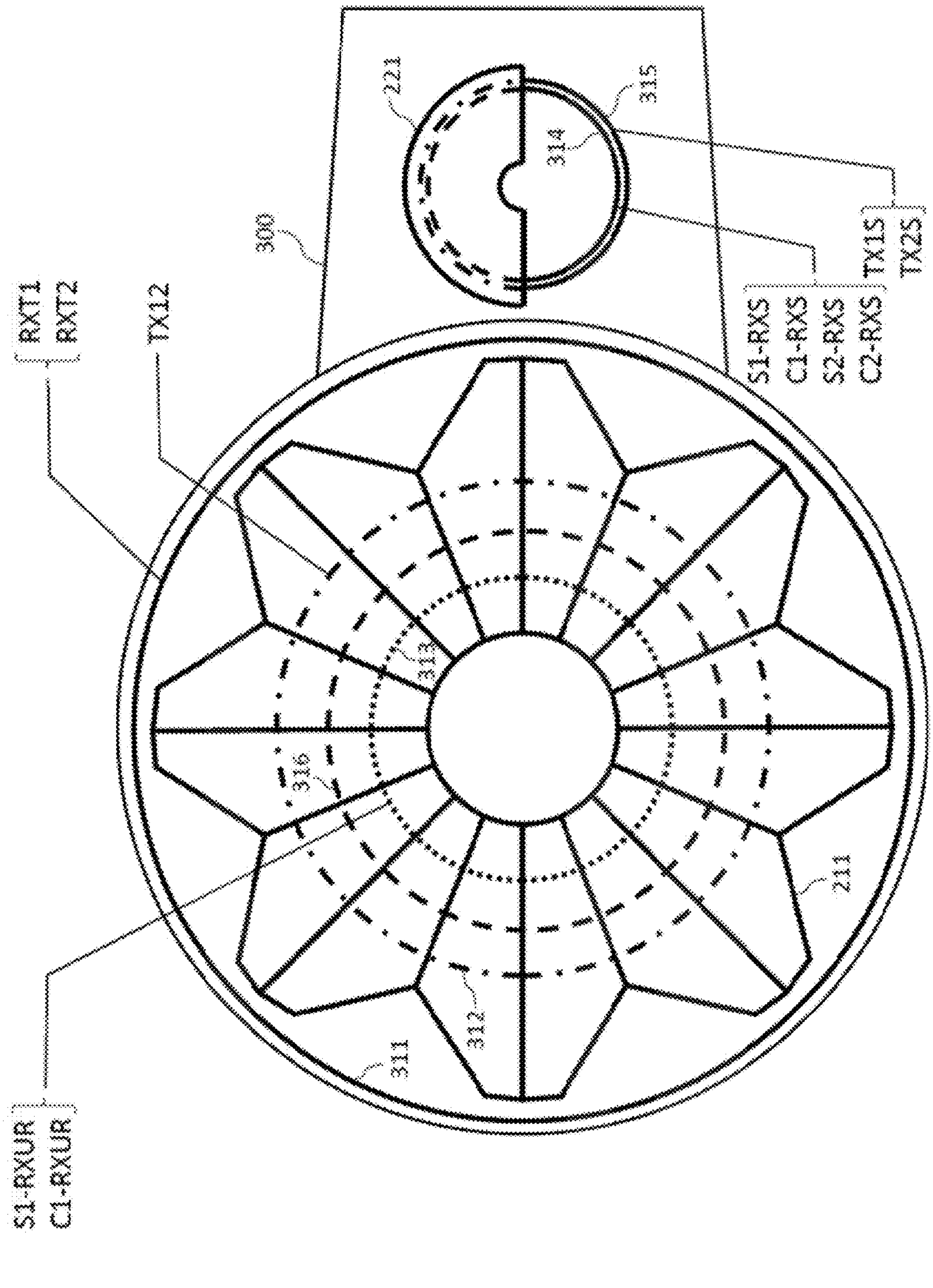

RXT1
RXT2
TX12
300
221
315
314
TX1S
TX2S
S1-RXS
C1-RXS
S2-RXS
C2-RXS
313
316
211
312
311
S1-RXUR
C1-RXUR

FIG. 2

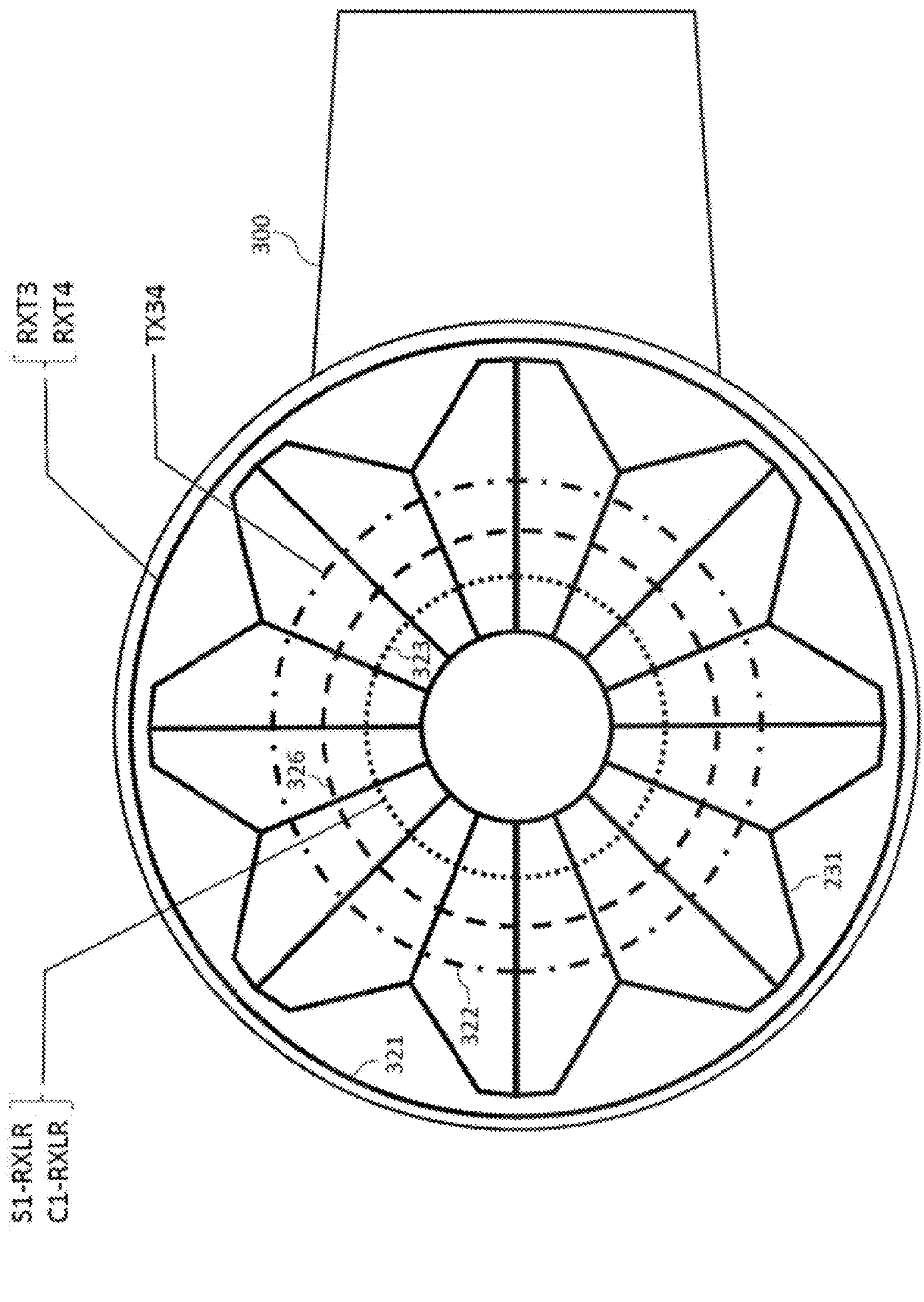


FIG. 3

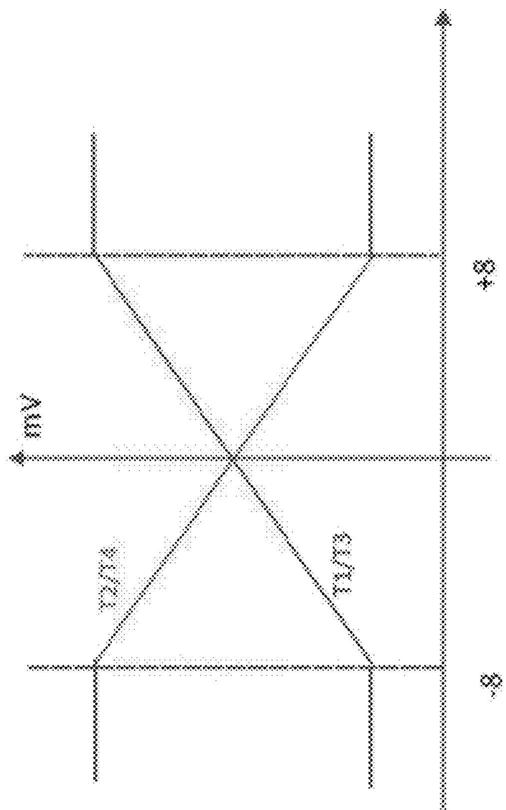

mV
T2/T4
T1/T3
-8
+8

FIG. 4A

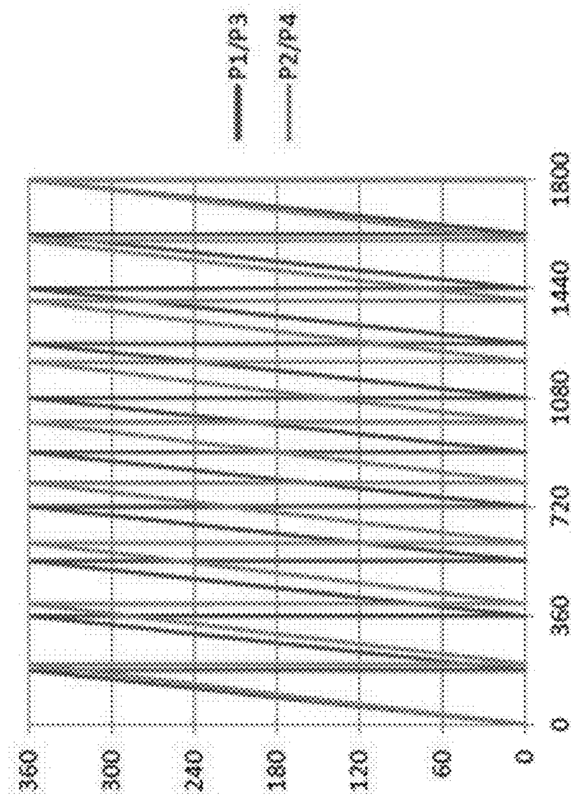

P1/P3
P2/P4
360
300
240
180
120
60
0
0
360
720
1080
1440
1800

FIG. 4B

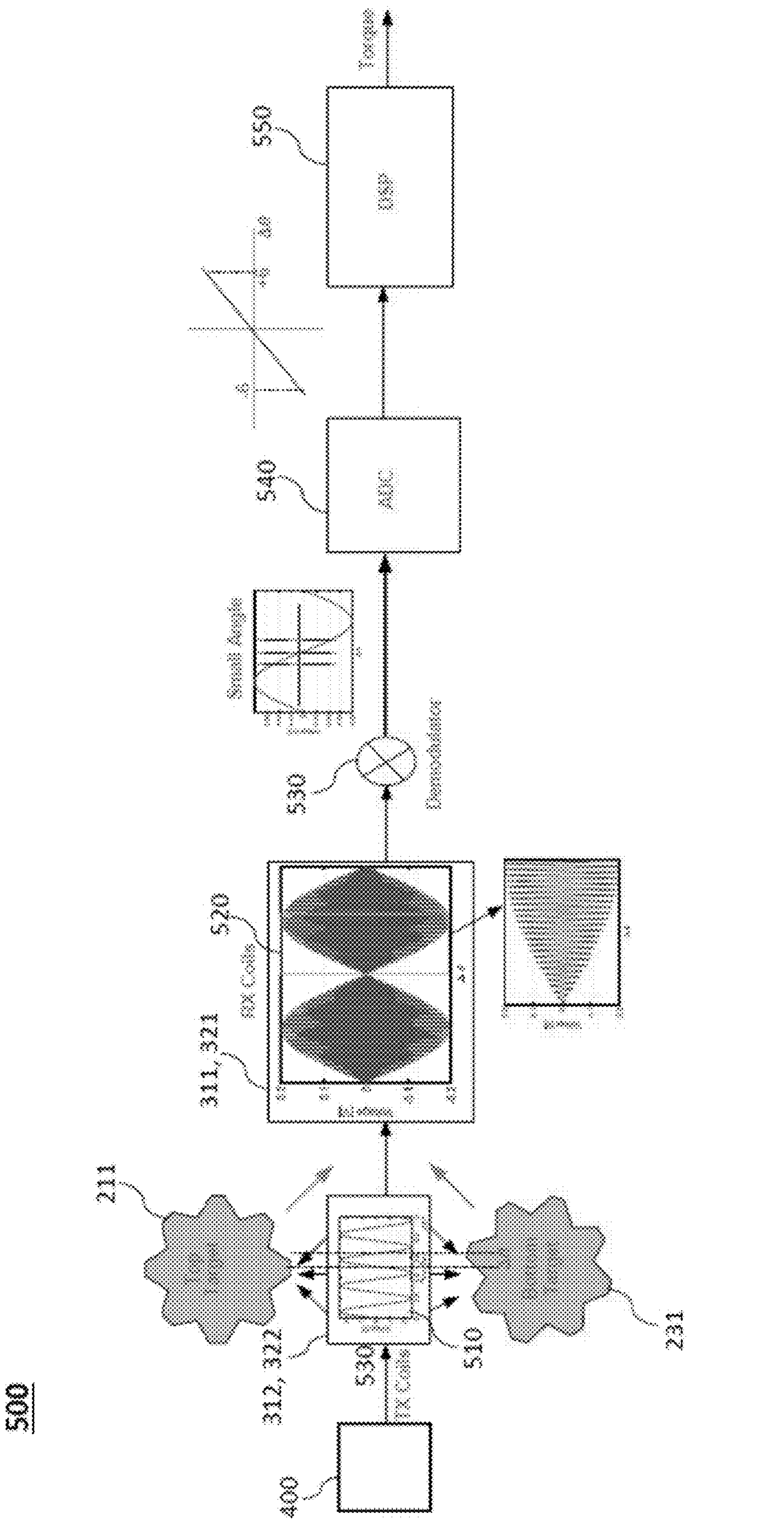
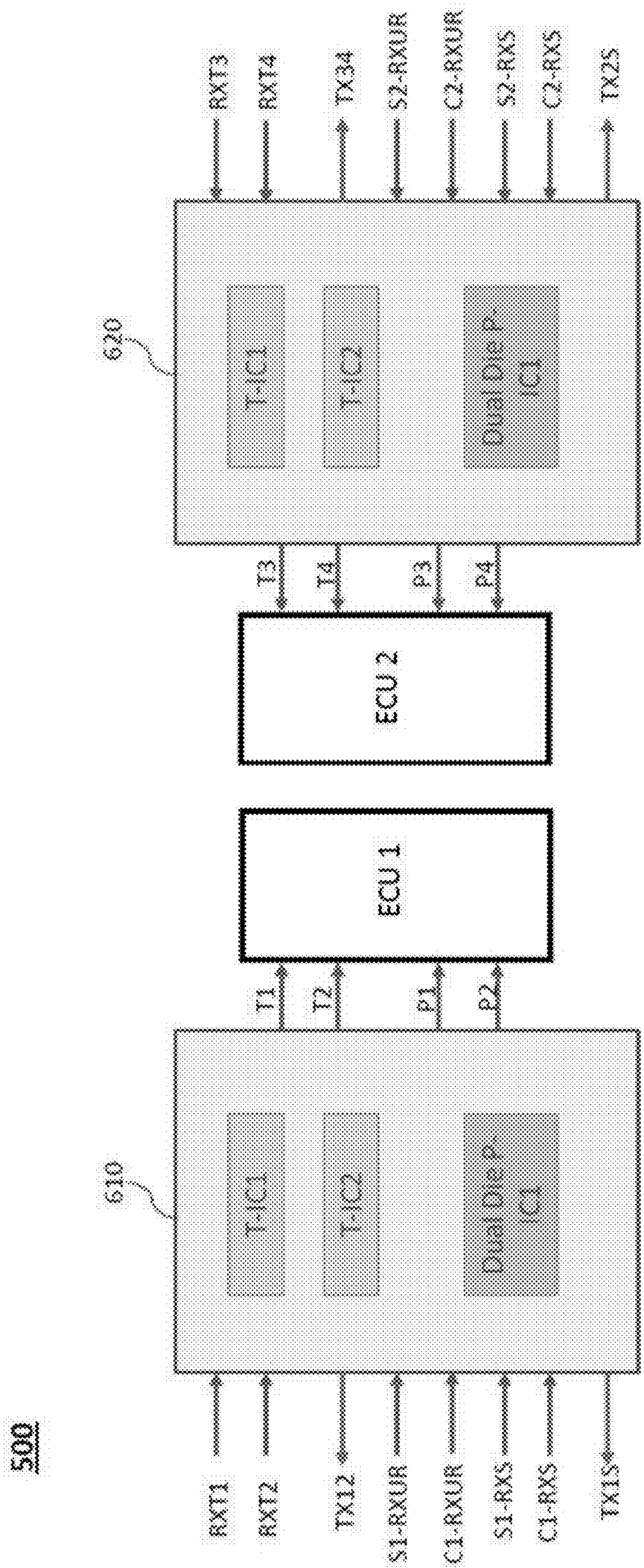


FIG. 5


500
610
T-IC1
T-IC2
Dual Die P-
IC1
RXT1
RXT2
TX12
S1-RXUR
C1-RXUR
S1-RXS
C1-RXS
TX1S
T1
T2
P1
P2
ECU 1
ECU 2
T3
T4
P3
P4
620
T-IC1
T-IC2
Dual Die P-
IC1
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RXT4
TX34
S2-RXUR
C2-RXUR
S2-RXS
C2-RXS
TX2S

FIG. 6

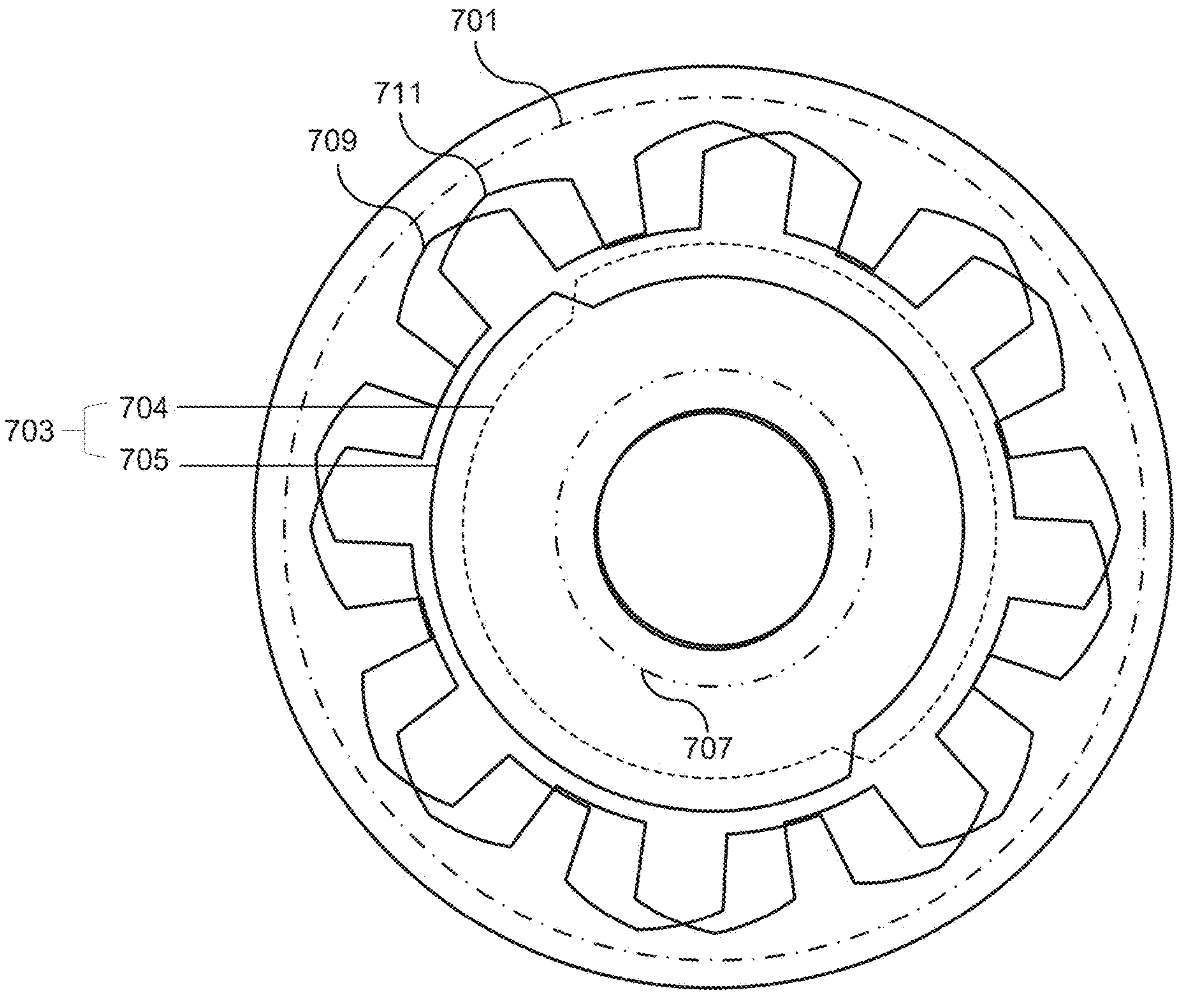

FIG. 7

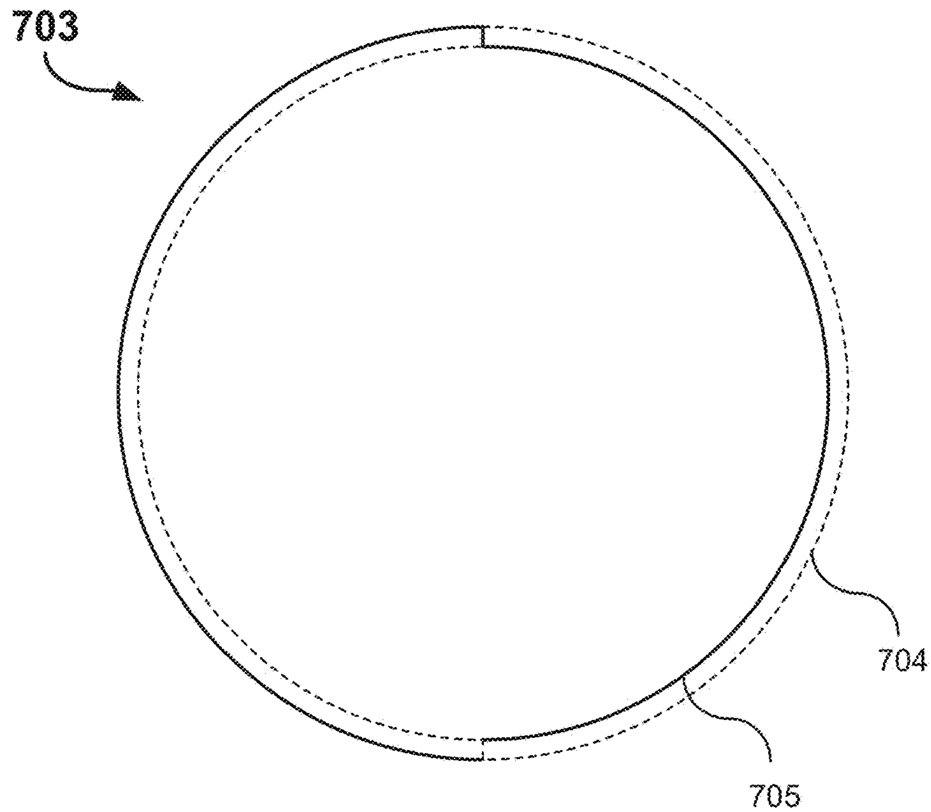

FIG. 8A

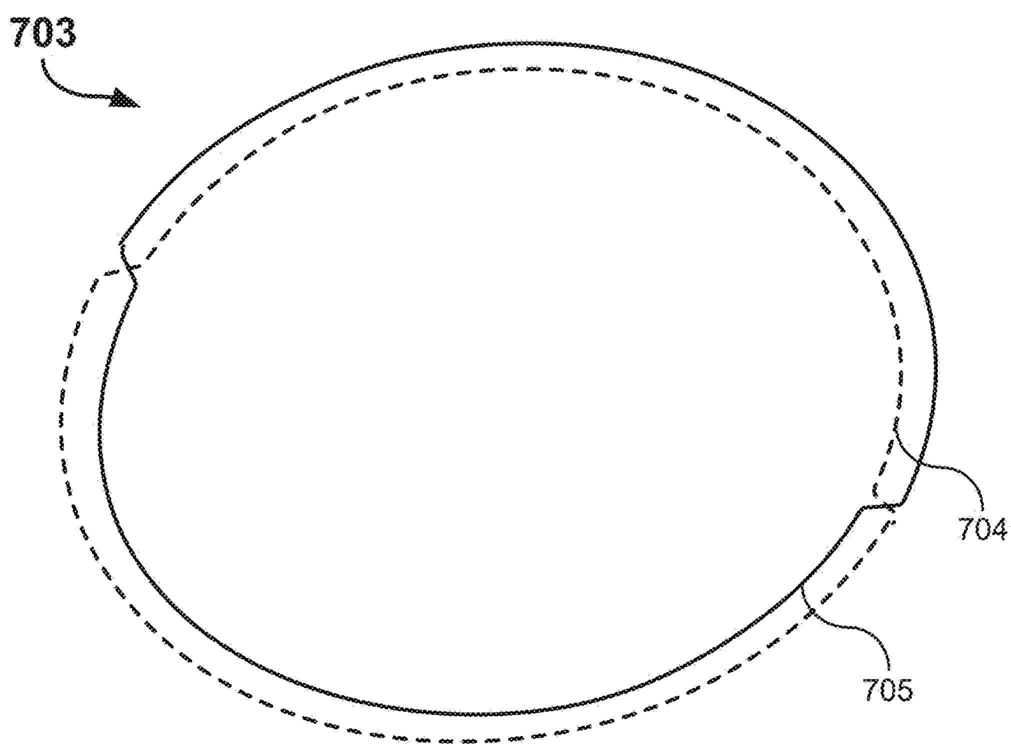

FIG. 8B

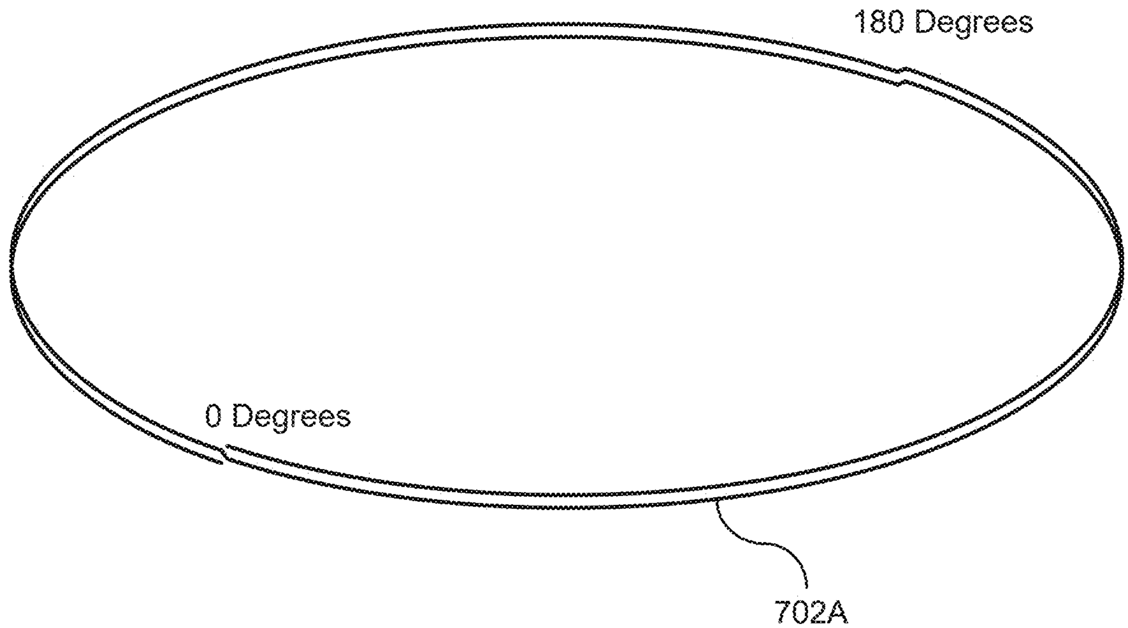

FIG. 9A

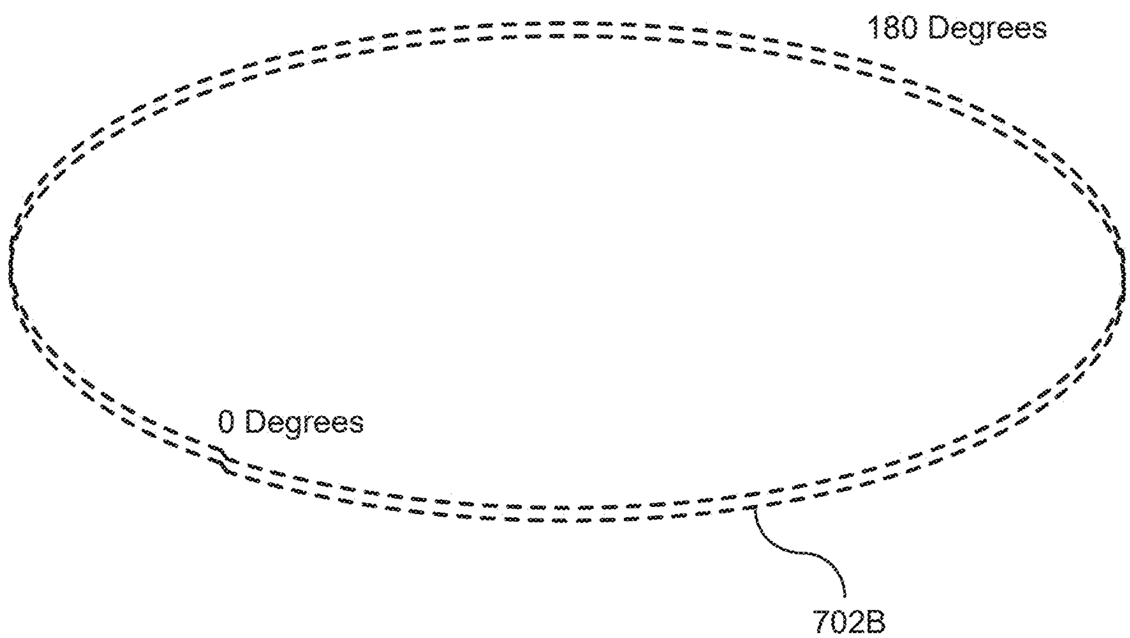

FIG. 9B

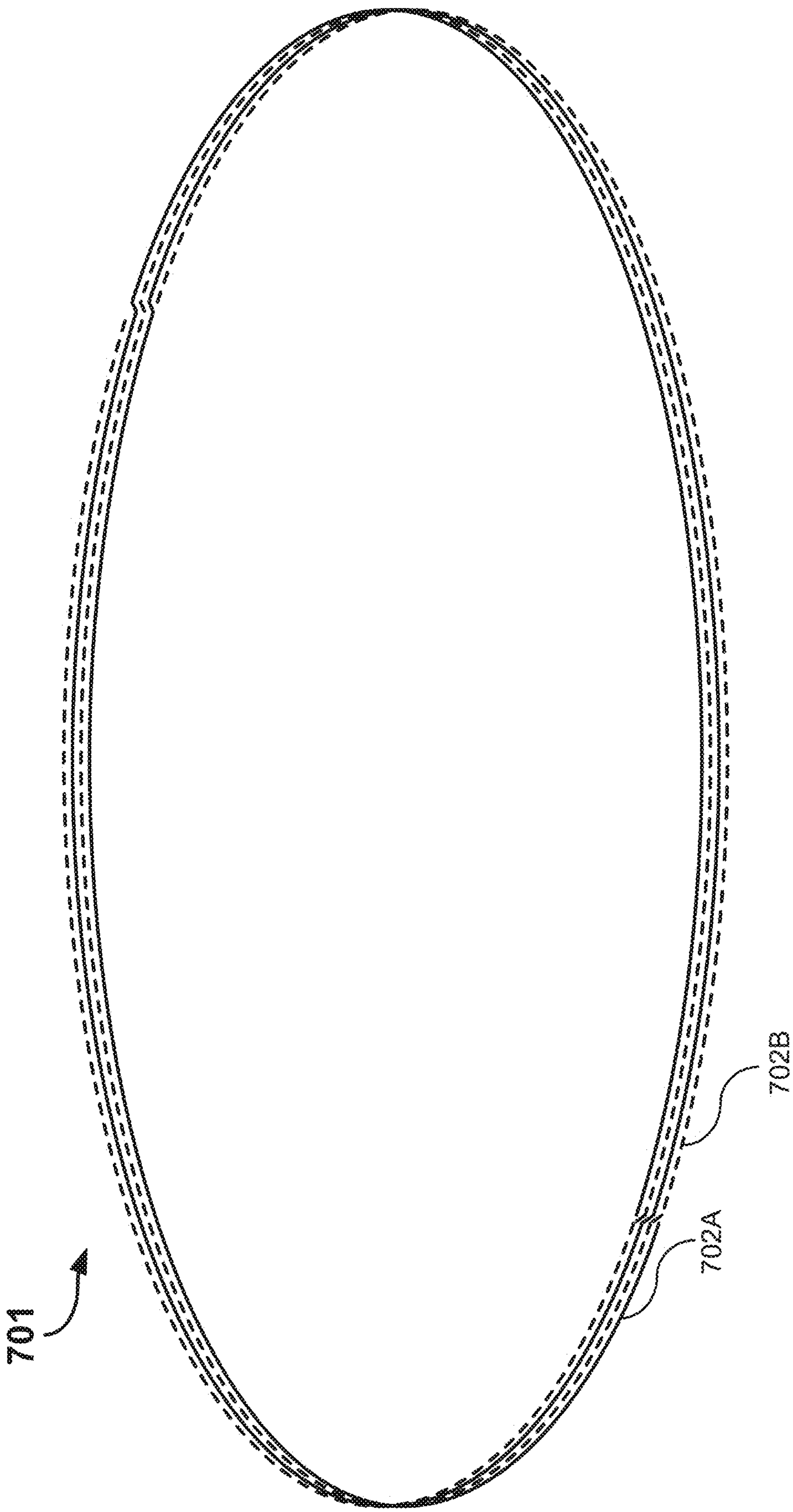

FIG. 9C

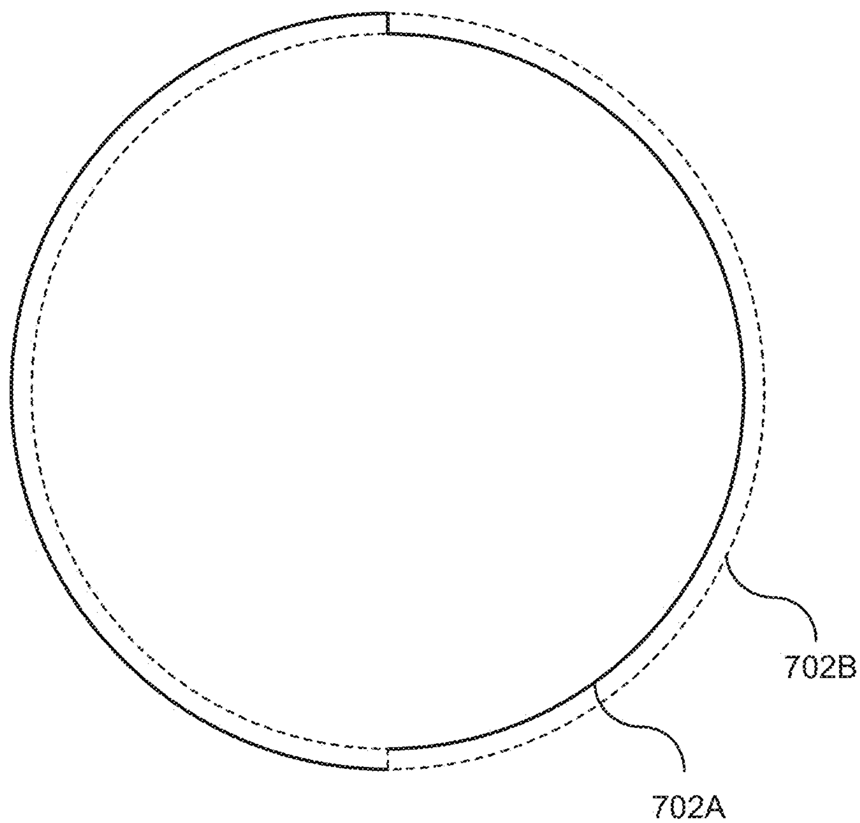

FIG. 9D

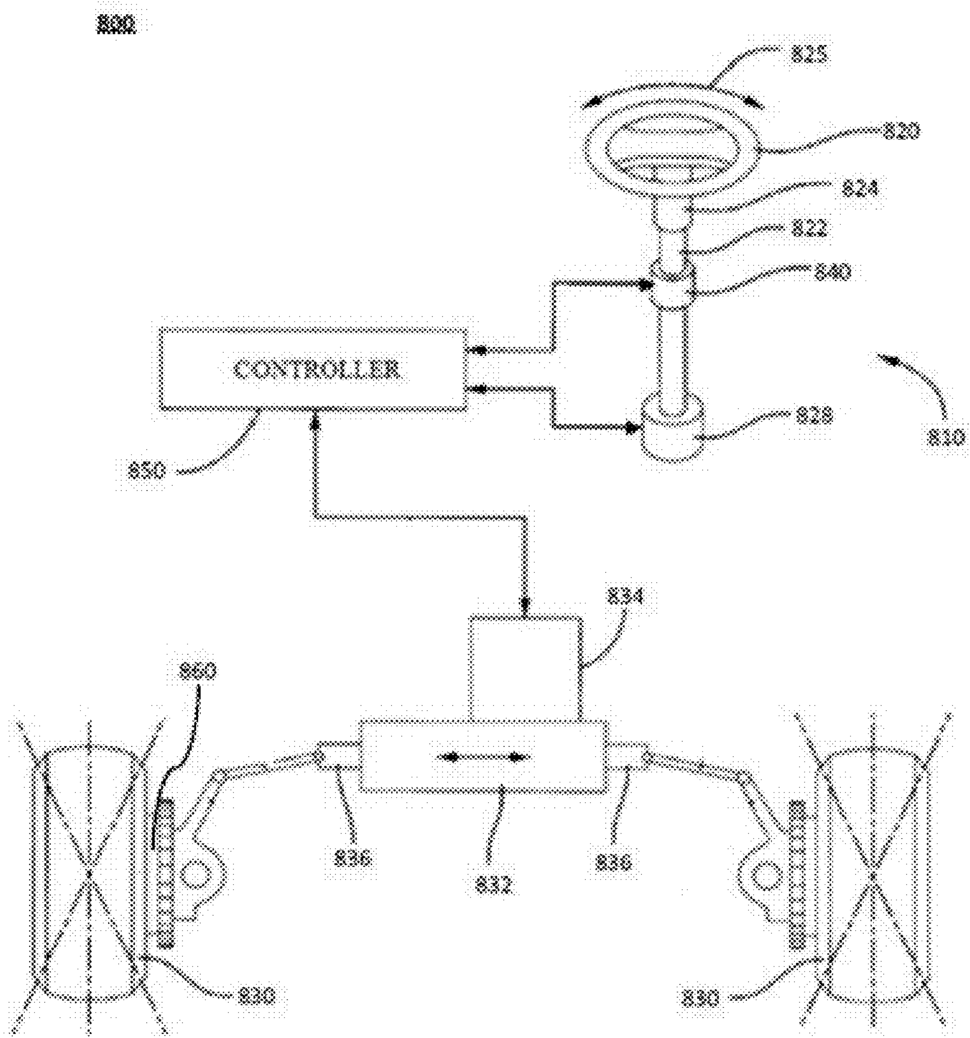
800
825
820
824
822
840
CONTROLLER
828
810
850
834
860
836
832
836
830
830

FIG. 10

COIL DESIGN FOR INDUCTIVE SENSOR SYSTEM COMPRISING INDUCTIVE TORQUE AND POSITION SENSOR ASSEMBLIES

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit from and the priority to U.S. Patent Application Ser. No. 63/562,229, filed on Mar. 6, 2024, titled "INDUCTIVE TORQUE & ABSOLUTE POSITION SENSOR", which is hereby incorporated herein by reference in its entirety.

BACKGROUND

[0002] The present disclosure generally relates to an inductive sensor system including inductive torque and position sensor assemblies. More specifically, some embodiments of the present disclosure relate to inductive torque and position sensor assemblies for a steering system of a vehicle by using electromagnetic principles such as inductance to determine torque applied to a steering wheel and a position of a steering shaft.

[0003] A steering system used in an automotive vehicle typically includes an input shaft connected to a steering wheel. The input shaft is then connected to an output shaft through a torsion bar and the output shaft, in turn, is mechanically connected through linkage to vehicle wheels. Consequently, the rotation of the steering wheel pivots the wheels of the automotive vehicle through the input shaft, torsion bar, output shaft, and steering linkage.

[0004] In many situations, it is highly desirable to determine the angular position of the input or output shaft and the angular deflection between the input shaft and the output shaft of the steering mechanism. The angular position of the input shaft may indicate where a driver wants to steer, matching the steering wheel with the vehicle wheels. And, the degree of angular deflection between the input shaft and the output shaft, i.e. the angular deflection of the torsion bar, is then utilized by a controller to detect the applied steering wheel torque and then to determine the appropriate amount of assist provided by the power steering for the vehicle.

[0005] In addition, there has been a recent trend towards electronically controlled steering systems, for instance, a steer-by-wire system which does not have a mechanical linkage between the steering wheel and the vehicle wheels. In the steer-by-wire system, the absolute position of the input shaft and the torque applied to the steering wheel can be used to electrically control the vehicle wheels.

SUMMARY

[0006] The features and advantages of the present disclosure will be more readily understood and apparent from the following detailed description, which should be read in conjunction with the accompanying drawings, and from the claims which are appended to the end of the detailed description.

[0007] According to some embodiments of the present disclosure, an inductive sensor system may comprise: an upper rotor comprising an upper target having a first metallic pattern; a lower rotor comprising a lower target having a second metallic pattern; and a stationary circuit board positioned between the upper rotor and the lower rotor, the circuit board comprising: one or more transmitter coil sets configured to generate electromagnetic field, one or more receiver coil sets for sensing relative angular displacement movement between the upper rotor and the lower rotor, wherein the one or more transmitter coil sets and the one or more receiver coil sets are circularly wound.

[0008] The one or more circularly wound receiver coil sets for sensing the relative angular displacement movement between the upper rotor and the lower rotor may be positioned radially outside the first metallic pattern of the upper target and the second metallic pattern of the lower target, and the one or more circularly wound transmitter coil sets may be positioned radially inside the first metallic pattern of the upper target and the second metallic pattern of the lower target.

[0009] The one or more circularly wound receiver coil sets for sensing the relative angular displacement movement between the upper rotor and the lower rotor may be positioned radially inside the first metallic pattern of the upper target and the second metallic pattern of the lower target, and the one or more circularly wound transmitter coil sets may be positioned radially inside the first metallic pattern of the upper target and the second metallic pattern of the lower target.

[0010] At least one of the one or more circularly wound receiver coil sets for sensing the relative angular displacement movement between the upper rotor and the lower rotor may be positioned radially outside the first metallic pattern of the upper target and the second metallic pattern of the lower target, and another or other of the one or more circularly wound receiver coil sets for sensing the relative angular displacement movement between the upper rotor and the lower rotor may be positioned radially inside the first metallic pattern of the upper target and the second metallic pattern of the lower target, and the one or more circularly wound transmitter coil sets may be positioned radially inside the first metallic pattern of the upper target and the second metallic pattern of the lower target, respectively.

[0011] The circuit board may comprise an other upper receiver coil set for sensing an angular position of the upper rotor.

[0012] The circuit board may comprise an other lower receiver coil set for sensing an angular position of the lower rotor.

[0013] The inductive sensor system may further comprise: an auxiliary rotor rotatably engaged with the upper rotor and having a third metallic pattern; and an auxiliary transmitter coil set and an auxiliary receiver coil set included in the circuit board or disposed on an upper surface of the circuit board.

[0014] The inductive sensor system may further comprise: an auxiliary rotor rotatably engaged with the lower rotor and having a third metallic pattern; and an auxiliary transmitter coil set and an auxiliary receiver coil set included in the circuit board or disposed on an lower surface of the circuit board.

[0015] The inductive sensor system may further comprise: an auxiliary rotor rotatably engaged with the upper or lower rotor and having magnetic material; and a sensor configured to sense magnetic field and positioned below or above the auxiliary rotor.

[0016] The upper receiver coil set and the other upper receiver coil set may be disposed on an upper surface of the printed circuit board, the lower receiver coil and the other lower receiver coil set may be disposed on a lower surface of the printed circuit board, and the one or more transmitter coil sets may be disposed on the upper surface of the circuit board, the lower surface of the circuit board, or inside the circuit board.

[0017] The circuit board may have multiple layers including upper layers and lower layers, the upper receiver coil set, and the other upper receiver coil set may be disposed on or between the upper layers of the circuit board, the lower receiver coil set, and the other lower receiver coil set may be disposed on or between the lower layers of the circuit board, and the one or more transmitter coil sets may be disposed on the upper or lower surface of the circuit board or between the upper surface and the lower surface of the circuit board.

[0018] The first metallic pattern of the first target and/or the second metallic pattern of the second target may have a plurality of circumferentially adjacent lobes.

[0019] The third metallic pattern of the second upper rotor may have a substantially half circular or polygonal shape.

[0020] The third metallic pattern of the second upper rotor may have a substantially half circular or polygonal shape.

[0021] The upper rotor and the auxiliary rotor may have gear teeth to be engaged with each other.

[0022] The lower rotor and the auxiliary rotor may have gear teeth to be engaged with each other.

[0023] A gear ratio between the upper rotor and the auxiliary rotor may be around from 1.8 to 2.7.

[0024] The upper rotor may be comprised in or coupled to an upper shaft coupled to a steering wheel, the lower rotor may be comprised in or coupled to a lower shaft, and a torsion bar may be coupled between the upper shaft and the lower shaft.

[0025] According to certain embodiments of the present disclosure, an inductive sensor system may comprise: an upper rotor comprising an upper target having a first metallic pattern; a lower rotor comprising a lower target having a second metallic pattern; an auxiliary rotor rotatably engaged with the upper rotor or the lower rotor; and a stationary circuit board positioned between the upper rotor and the lower rotor, the circuit board comprising: one or more transmitter coil sets configured to generate electromagnetic field, one or more receiver coil sets for sensing relative angular displacement movement between the upper rotor and the lower rotor, and one or more receiver coil sets for sensing an angular position of the upper rotor and/or the lower rotor, wherein: the one or more transmitter coil sets and the one or more receiver coil sets for sensing the relative angular displacement movement between the upper rotor and the lower rotor are circularly wound.

[0026] The auxiliary rotor has a third metallic pattern, and an auxiliary transmitter coil set and an auxiliary receiver coil set included in the circuit board or disposed on a surface of the circuit board.

[0027] The inductive sensor system may further comprise a sensor configured to sense magnetic field and positioned below or above the auxiliary rotor, wherein the auxiliary rotor rotatably engaged with the upper or lower rotor includes magnetic material.

[0028] According to some embodiments of the present disclosure, an inductive sensor system may comprise: a circuit board. The circuit board may comprise: a transmitter coil set comprising at least: a first transmitter coil comprising a first portion with a first radius and a second portion with a second radius smaller than the first radius; and a second transmitter coil comprising a first portion with a third radius identical to the first radius and a second portion with a fourth radius identical to the second radius, wherein the first transmitter coil and the second transmitter coil are circularly wound in a manner where the first radius and the fourth radius are on a first side of the circuit board while the second radius and the third radius are on a second side of the circuit board. The circuit board may further comprise: a receiver coil set comprising at least a first receiver coil and a second receiver coil, wherein the first receiver coil and the second receiver coil are circularly wound.

[0029] The receiver coil set surrounds the transmitter coil set, the circuit board comprises at least a first surface and a second surface opposite to the first surface, the first surface on the first side of the circuit board comprises the first portion of the first transmitter coil, the first surface on the second side of the circuit board comprises the first portion of the second transmitter coil, the second surface on the first side of the circuit board comprises the second portion of the second transmitter coil, and the second surface on the second side of the circuit board comprises the second portion of the first transmitter coil.

[0030] The first transmitter coil further comprises a third portion where the first radius transitions into the second radius and a fourth portion where the second radius transitions back into the first radius, the second transmitter coil further comprises a third portion where the third radius transitions into the fourth radius and a fourth portion where the fourth radius transitions back into the third radius, and the third portion of the first transmitter coil overlaps with the third portion of the second transmitter coil and the fourth portion of the first transmitter coil overlaps with the fourth portion of the second transmitter coil within a body of the circuit board that separates the first surface from the second surface.

[0031] The inductive sensor system may further include: an upper rotor comprising an upper target having a first metallic pattern; and a lower rotor comprising a lower target having a second metallic pattern, wherein the circuit board is a stationary circuit board that is positioned between the upper rotor and the lower rotor, the transmitter coil set is configured to generate an electromagnetic field, and the receiver coil set is configured to sense relative angular displacement movement between the upper rotor and the lower rotor.

[0032] The circularly wound receiver coil set is positioned radially outside an outer most surface of the first metallic pattern and of the second metallic pattern, and the circularly wound transmitter coil set is positioned radially within the outer most surface of the first metallic pattern and of the second metallic pattern.

[0033] The first metallic pattern of the upper target and/or the second metallic pattern of the lower target have a plurality of circumferentially adjacent lobes.

[0034] The circuit board may further include: a bias coil set comprising at least a first bias coil and a second bias coil, wherein the first bias coil and the second bias coil are circularly wound and the bias coil set is surrounded by the transmitter coil set.

[0035] The circuit board comprises N number of layers, each of the first receiver coil and the second receiver coil comprises a first coiled structure comprising N divided by 2 number of full turns, and each of the layers of the circuit board comprises at least a half turn of each of the first receiver coil and the second receiver coil.

[0036] The half turn of the first receiver coil and of the second receiver coil on a same layer of the layers comprise a same radius.

[0037] The half turn of the first receiver coil and of the second receiver coil on a same layer of the layers comprise different radiuses.

[0038] Each of the first bias coil and the second bias coil comprises a second coiled structure comprising N divided by 2 number of full turns, and each of the layers of the circuit board comprises at least a half turn of each of the first bias coil and the second bias coil.

[0039] The half turn of each of the first bias coil and the second bias coil on a same layer of the layers comprise a same radius.

[0040] The half turn of each of the first bias coil and the second bias coil on a same layer of the layers comprise different radiuses.

[0041] A total first number of the full turns of the receiver coil set is identical to a second total number of the full turns of the bias coil set.

[0042] The bias coil set is disposed in series with the receiver coil set and is configured to cancel DC bias generated by the receiver coil set.

[0043] According to some embodiments of the present disclosure, a motor vehicle may comprise: one or more road wheels; a steering wheel coupled to at least one of the one or more road wheels; and an inductive sensor system. The inductive sensor system may comprise: a circuit board. The circuit board may comprise: a transmitter coil set comprising at least: a first transmitter coil comprising a first portion with a first radius and a second portion with a second radius smaller than the first radius; and a second transmitter coil comprising a first portion with a third radius identical to the first radius and a second portion with a fourth radius identical to the second radius, wherein the first transmitter coil and the second transmitter coil are circularly wound in a manner where the first radius and the fourth radius are on a first side of the circuit board while the second radius and the third radius are on a second side of the circuit board; and a receiver coil set comprising at least a first receiver coil and a second receiver coil, wherein the first receiver coil and the second receiver coil are circularly wound.

[0044] The receiver coil set surrounds the transmitter coil set, the circuit board comprises at least a first surface and a second surface opposite to the first surface, the first surface on the first side of the circuit board comprises the first portion of the first transmitter coil, the first surface on the second side of the circuit board comprises the first portion of the second transmitter coil, the second surface on the first side of the circuit board comprises the second portion of the second transmitter coil, and the second surface on the second side of the circuit board comprises the second portion of the first transmitter coil.

[0045] The first transmitter coil further comprises a third portion where the first radius transitions into the second radius and a fourth portion where the second radius transitions back into the first radius, the second transmitter coil further comprises a third portion where the third radius transitions into the fourth radius and a fourth portion where the fourth radius transitions back into the third radius, wherein the third portion of the first transmitter coil overlaps with the third portion of the second transmitter coil and the fourth portion of the first transmitter coil overlaps with the fourth portion of the second transmitter coil within a body of the circuit board that separates the first surface from the second surface.

[0046] The inductive sensor system may further comprise: an upper rotor comprising an upper target having a first metallic pattern; and a lower rotor comprising a lower target having a second metallic pattern, wherein the circuit board is a stationary circuit board that is positioned between the upper rotor and the lower rotor, the transmitter coil set is configured to generate an electromagnetic field, and the receiver coil set is configured to sense relative angular displacement movement between the upper rotor and the lower rotor.

[0047] The upper rotor is comprised in or coupled to an upper shaft coupled to the steering wheel, the lower rotor is comprised in or coupled to a lower shaft, and a torsion bar is coupled between the upper shaft and the lower shaft.

[0048] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0049] Various embodiments in accordance with the present disclosure will be described with reference to the drawings, in which:

[0050] FIG. **1** is a cross-sectional view of a steering column having an inductive sensor system according to an embodiment of the present disclosure.

[0051] FIG. **2** is a top view of an inductive sensor system according to an embodiment of the present disclosure.

[0052] FIG. **3** is a bottom view of an inductive sensor system according to an embodiment of the present disclosure.

[0053] FIG. **4**A is a graph for showing linear torque signals generated by an inductive torque assembly of an inductive sensor system according to an embodiment of the present disclosure.

[0054] FIG. **4**B is a graph for showing output signals of primary and auxiliary position sensor assemblies of an inductive sensor system according to an embodiment of the present disclosure.

[0055] FIG. **5** is a conceptual diagram for illustrating a controller and a process of detecting a torque applied to a steering wheel according to an embodiment of the present disclosure.

[0056] FIG. **6** is a block diagram of a controller according to an embodiment of the present disclosure.

[0057] FIG. **7** shows a transparent top-down view of the inductive sensor system according to an embodiment of the present disclosure.

[0058] FIG. **8**A shows a top view of the transmitter coil set according to an embodiment of the present disclosure.

[0059] FIG. **8**B shows a perspective view of the transmitter coil set according to an embodiment of the present disclosure.

[0060] FIGS. **9**A-**9**D shows perspective views of a receiver coil set of the inductive sensor system according to an embodiment of the present disclosure.

[0061] FIG. **10** is a schematic view of a vehicle including a steering system and a brake assembly according to an exemplary embodiment of the present disclosure.

[0062] Corresponding numerals and symbols in the different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the embodiments and are not necessarily drawn to scale.

DETAILED DESCRIPTION OF EMBODIMENTS

[0063] In the following detailed description, reference is made to the accompanying drawings which form a part of the present disclosure, and in which are shown by way of illustration specific embodiments in which the invention may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the invention, and it is to be understood that other embodiments may be utilized and that structural, logical and electrical changes may be made without departing from the spirit and scope of the invention. The following detailed description is therefore not to be taken in a limiting sense, and the scope of the invention is defined only by the appended claims and equivalents thereof. Like numbers in the figures refer to like components, which should be apparent from the context of use.

[0064] FIG. **1** is a cross-sectional view of a steering column having an inductive sensor system according to an embodiment of the present disclosure.

[0065] The inductive sensor system according an embodiment of the present disclosure may comprise a torque sensor assembly and an angle sensor assembly. The torque sensor assembly is required for information about torque applied to a steering wheel which is proportion to a relative position between an upper shaft and a lower shaft. The angle sensor assembly is required for absolute position information of the upper shaft or the lower shaft. The angle sensor assembly provides an output signal that is proportional to the rotation angle of the upper shaft or the lower shaft.

[0066] A vehicle (see e.g., FIG. **10**) has a steering column **100** includes an upper shaft (or an input shaft) **110** and a lower shaft (or an output shaft) **120**. The upper shaft **110** may be mechanically connected or fixed to a steering wheel **105** and the lower shaft **120** may be mechanically connected to vehicle wheels in a conventional mechanical steering system or a feedback actuator (e.g. an electric motor) in a steer-by-wire steering system. The upper shaft **110** and the lower shaft **120** may be axially aligned with each other.

[0067] The upper shaft **110** and the lower shaft **120** are connected by a torsion bar or beam **130**. The torsion bar **130** may be configured to allow the upper shaft **110** and the lower shaft **120** to rotate slightly relative to each other in response to torque applied to the steering wheel **105**.

[0068] An upper rotor **210** is fixedly coupled to the upper shaft **110** or is a part of the upper shaft **110**. The upper rotor **210** is configured to be rotatable together with the upper shaft **110**. For example, the upper rotor **210** may be a floating printed circuit board (PCB).

[0069] A lower rotor **230** is fixedly coupled to the lower shaft **120** or is a part of the lower shaft **120**. The lower rotor **230** is configured to be rotatable together with the lower shaft **120**. For example, the lower rotor **230** may be a floating PCB.

[0070] A stator **300** (e.g. a stationary circuit board) may be positioned between the upper rotor **210** and the lower rotor **230**. The stator **300** is coaxially mounted around the steering column **100**. For example, the stator **300** may be adjacent around the torsion bar **130**. Alternatively, the stator **300** may be located adjacent around the upper rotor **210** or the lower rotor **230**. The stator **300** may be fixed by being directly or indirectly coupled to a vehicle body. Accordingly, the stator **300** does not move relative to the steering column **100**, while the upper rotor **210** can rotate with the upper shaft **110** and the lower rotor **230** can rotate with the lower shaft **120** relative to the stator **300**. The stator **300** may be arranged to be parallel to the upper rotor **210** and/or the lower rotor **230**.

[0071] An oscillator **400** illustrated in FIG. **5** may be configured to oscillate at a high frequency, for example, but not limited to, 2 to 4 MHz. The oscillator **400** may be electrically connected to one or more excitation or transmitter coil sets **312** and/or **322**, and an auxiliary excitation or transmitter coil set **315** to excite one or more relative angular displacement receiver coil sets **311** and **321**, an upper angular position receiver coil set **313**, a lower angular position receiver coil set **323**, and an auxiliary angular position receiver coil set **314**.

[0072] One or more excitation or transmitter coil set **312** and/or **322** are included in the stator **30** and/or disposed on an upper and/or lower surface of the stator **300**. For example, the excitation or transmitter coil set **312** and/or **322** may be formed by conductive traces on the upper or lower surface of the stator **300** or electrically conductive pathways on a multi-layer PCB of the stator **300**. As an example, at least a part of one coil of the excitation or transmitter coil set **312** and/or **322** is placed on one layer of the multi-layer PCB of the stator **300**, and at least a part of another coil of the excitation or transmitter coil set **312** and/or **322** is placed on another layer of the multi-layer PCB of the stator **300**. The excitation or transmitter coil set **312** and/or **322** is electronically connected to the oscillator **400**. The excitation or transmitter coil set **312** and/or **322** generates an electromagnetic field over an upper target **211** of the upper rotor **210** and the lower target **231** of the lower rotor **230** by a radio-frequency signal generated by the oscillator **400**. In FIG. **1**, one excitation or transmitter coil set **312** for one transmittal channel is disposed on the upper surface of the stator **300** and the other excitation or transmitter coil set **322** for another transmittal channel is disposed on the lower surface of the stator **300**. However, an excitation or transmitter coil set can be positioned on either one of the upper surface of the stator **300** or the lower surface of the stator **300**. Alternatively, one or more excitation or transmitter coil sets may be positioned between the multiple layers of the multi-layer PCB of the stator **300**.

[0073] The upper target **211** may be included in or attached to the upper rotor **210**. The upper target **211** may be an electrically conductive coupler. The upper target **211** may be placed in proximity to the excitation or transmitter coil set **312** and/or **322**. The upper target **211** may have a first metallic pattern. For instance, the upper target **221** can include a closed conductive loop or multiple conductive loops. The upper target **211** may have, for example, but not limited to, a multi lobe shape having the plurality of circumferentially adjacent lobes. The upper target **211** can be configured to affect the electromagnetic field generated by the excitation or transmitter coil set **312** and/or **322**.

[0074] The lower target **231** may be included in or attached to the lower rotor **230**. The lower target **231** may be an electrically conductive coupler. The lower target **231** may be placed in proximity to the excitation or transmitter coil set **322** and/or **312**. The lower target **231** may have a second metallic pattern. For instance, the lower target **231** can include a closed conductive loop or multiple conductive loops. The lower target **231** may have, for example, but not limited to, a multi lobe shape having the plurality of circumferentially adjacent lobes. The second metallic pattern of the lower target **231** may be identical or different to or from the first metallic pattern of the upper target **211**. The lower target **231** can be configured to affect the electromagnetic field generated by the excitation or transmitter coil set **312** and/or **322**.

[0075] One or more relative angular displacement receiver coil sets **311** and **321** for sensing relative angular displacement movement between the upper rotor **210** and the lower rotor **230** are included in or disposed on an upper and/or lower surface of the stator **300**. For example, the relative angular displacement receiver coil sets **311** and/or **321** may be formed by conductive traces on the upper and/or lower surface of the stator **300** or electrically conductive pathways on a multi-layer PCB of the stator **300**. The relative angular displacement receiver coil set **311** and **321** may be placed in proximity to the upper target **211** and the lower target **231** and positioned within the electromagnetic fields generated by the transmitter coil set **312** and/or **322**. The relative angular displacement receiver coil set **311** and **321** may be configured to generate a signal (e.g. voltage or current) in response to induction by the electromagnetic fields generated by the transmitter coil set **312** and **322** and altered by the upper target **211** and the lower target **231**. The relative angular displacement receiver coil set **311** and/or **321** is electrically connected to a controller **500**, illustrated in FIGS. **5** and **6**, to output the signal (e.g. voltage or current) to the controller **500**.

[0076] The relative angular displacement receiver coil sets **311** and/or **321** for sensing relative angular displacement movement between the upper rotor **210** and the lower rotor **230** may be positioned radially outside the metallic pattern of the upper target **211** and the lower target **231**. The relative angular displacement receiver coil sets **311** and/or **321** are circularly wound. A winding diameter of the relative angular displacement receiver coil set **311** and/or **321** for sensing the relative angular displacement movement between the upper rotor **210** and the lower rotor **230** is greater than a winding diameter of the excitation or transmitter coil **312** and/or **322**. The relative angular displacement receiver coil set **311** and/or **321** for sensing the relative angular displacement movement between the upper rotor **210** and the lower rotor **230** may surround the excitation or transmitter coil set **312** and/or **322**. By arranging the relative angular displacement receiver coil set **311** and/or **321** for sensing relative angular displacement movement between the upper rotor **210** and the lower rotor **230** radially outside the metallic pattern of the upper target **211** and/or the lower target **231**, the rotational accuracy for sensing the torque applied to the steering column **105** such as the relative angular displacement movement between the upper rotor **210** and the lower rotor **230** can be improved.

[0077] Alternatively, the relative angular displacement receiver coil sets **311** and/or **321** for sensing relative angular displacement movement between the upper rotor **210** and the lower rotor **230** may be positioned radially inside the metallic pattern of the upper target **211** and the lower target **231**. Or, one or more of the relative angular displacement receiver coil sets **311** and/or **321** for sensing relative angular displacement movement between the upper rotor **210** and the lower rotor **230** may be positioned radially inside the metallic pattern of the upper target **211** and the lower target **231**, while remaining another or other of the relative angular displacement receiver coil sets **311** and/or **321** for sensing relative angular displacement movement between the upper rotor **210** and the lower rotor **230** may be positioned radially outside the metallic pattern of the upper target **211** and the lower target **231**.

[0078] A reference signal can be determined from a combination of receiver signals, substantially independent of angular positions of the upper target **211** of the upper rotor **210** and angular positions of the lower target **231** of the lower rotor **230**, and this may be used to determine the number of rotations. Alternatively, a separate reference coil set **316** and/or **326** may be included in the stator **30** or disposed on an upper and/or lower surface of the stator **300**. For example, the reference coil set **316** and/or **326** may be included in the stator **300** to provide a reference signal. The reference coil set **316** and/or **326** may be formed by conductive traces on the upper and/or lower surface of the stator **300** or electrically conductive pathways on a multi-layer PCB of the stator **300**. The reference coil set **316** and/or **326** may have a similar configuration to the relative angular displacement receiver coil set **311** and/or **321**, but can be configured in such a way that a reference current or voltage induced in the reference coil by the transmitter coil is substantially independent of the position of the upper target **211** of the upper rotor **210** and the lower target **231** of the lower rotor **230**. The angular position or rotation of the upper target **211** of the upper rotor **210** and the lower target **231** of the lower rotor **230** does not affect the voltage or current induced into the reference coil set **316** and/or **326**. However, common mode signals such as electromagnetic interference, variations in exciter voltage, variations produced by temperature changes, and variations in the gap between the upper target **211** of the upper rotor **210** and the stator **300** and the gap between the lower target **231** of the lower rotor **230** and the stator **300**, will affect the voltage or current induced in the reference coil set **316** and/or **326** in the same way that they affect the voltage or current induced in the relative angular displacement receiver coil set **311** and/or **321**. By using a difference or ratio of the output signal of the relative angular displacement receiver coil set **311** and/or **321** and the output signal of the reference coil set **316** and/or **326**, the effects of the common mode factors can be suppressed. The reference coil set **316** and/or **326** may be circularly wound. A winding diameter of the reference coil set **316** and/or **326** may be smaller than both a winding diameter of the relative angular displacement receiver coil set **311** and/or **321** for sensing the relative angular displacement movement between the upper rotor **210** and the lower rotor **230** and a winding diameter of the excitation or transmitter coil set **312** and/or **322** in order to minimize the effect from the electromagnet fields associated with the excitation or transmitter coil **312** and/or **322** and the upper or lower target **211** or **231** of the upper or lower rotor **210** or **230**.

[0079] A torque determination may be made based on output signals of the relative angular displacement receiver coil set **311** and/or **321**. The output signals such as output voltages or currents of the relative angular displacement receiver coil set **311** and/or **321** can be used for sensing the relative angular displacement movement between the upper rotor **210** and the lower rotor **230**. The relative angular displacement movement between the upper rotor **210** and the lower rotor **230** is directly related to the torque or torsion applied to the steering wheel **105**. For example, by having the circularly wound relative angular displacement receiver coil set **311** and/or **321**, the output signal of the relative angular displacement receiver coil set **311** and/or **321** can be processed to provide a single linear signal over the torque applied the steering wheel **105** as illustrated in FIG. **4**A. An exemplary embodiment of a process for generating a single linear signal over the torque applied the steering wheel **105** will be described later with reference to FIG. **5**.

[0080] Since each of the relative angular displacement receiver coil set **311** and/or **321** includes an even number of oppositely wound loops, the output voltage on the relative angular displacement receiver coil set **311** and/or **321** may be indicative of a zero deflection between the upper shaft **110** and the lower shaft **120**, while a positive voltage may be indicative of torque in one direction between the upper shaft **110** and the lower shaft **120** and a negative voltage may be indicative of torque in the other direction between the upper shaft **110** and the lower shaft **120**.

[0081] FIG. **5** is a conceptual diagram for illustrating a controller and a process of detecting a torque applied to a steering wheel according to an embodiment of the present disclosure.

[0082] The controller **500** may include an electronic circuit such as an ASIC. The controller **500** is configured as a micro-processor configured to execute non-transient computer executable, instructions that are suitably stored on firmware, software, or otherwise for use in performing functions. Ends of the relative angular displacement receiver coil set **311** and/or **321** and the reference coil set **316** and **326** are connected to the controller **500** to process their output signals. The controller **500** may have a processor programmed to output the magnitude and direction of the relative angular displacement between the upper shaft **110** and the lower shaft **120** and the absolute rotational position of the upper shaft **110** and/or the lower shaft **120**.

[0083] The oscillator **400** is connected to the ends of the excitation or transmitter coil set **312** and/or **322**. The oscillator **400** provides excitation signals **510** such as alternating currents to the excitation or transmitter coil set **312** and/or **322**, thereby generating an alternating electromagnetic field, which subsequently induces signals in the excitation or transmitter coil set **312** and/or **322** through inductive coupling. The inductive coupling between the excitation or transmitter coil sets **312** and **322** and the receiver coil sets **311** and **321** is changed (e.g. reduced) by the targets **211** and **231** of the rotors **210** and **230**. However, the inductive coupling between the excitation or transmitter coil sets **312** and **322** and the reference coil sets **316** and **326** is not sensitive to the angular position of the targets **211** and **231** of the rotors **210** and **230**. In contrast, the output signals **520** of the receiver coil sets **311** and **321** are sensitive to the angular position of the targets **211** and **231** of the rotors **210** and **230**, so that a ratio of the output signals **520** of the receiver coil sets **311** and **321** and the output signals of the reference coil sets **316** and **326** is correlated with the angular position of the targets **211** and **231** of the rotors **210** and **230** while also being corrected for common mode factors as discussed above.

[0084] A demodulator **530** demodulates the output signal **520** combined by the output signal of the receiver coil sets **311** and **321** and the output signal of the reference coil sets **316** and **326**, an analog-to-digital converter (ADC) **540** converts the demodulated output signal to an analog signal, and a digital signal processor (DSP) **550** processes the converted analog signal to output an output signal indicative of the torque applied to the steering wheel **105**. The output signal indicative of the torque applied to the steering wheel **105** may be a linear output voltage as a function of angular displacement between the upper rotor **210** and the lower rotor **230** as illustrated in FIG. **4**A.

[0085] However, the relative angular displacement receiver coil set **311** and/or **321** cannot provide an absolute angular rotational position of the upper rotor **210** and the lower rotor **230**.

[0086] In order to determine the absolute angular rotational position of the upper rotor **210** and the lower rotor **230**, an auxiliary or satellite rotor **220** may be further included.

[0087] The auxiliary or satellite rotor **220** may be rotatably engaged with the upper rotor **210**. For instance, the upper rotor **210** and the auxiliary or satellite rotor **220** may have gear teeth meshed with each other. The number of teeth of the upper rotor **210** is different from the number of the auxiliary or satellite rotor **220** so that the upper rotor **210** and the auxiliary or satellite rotor **220** rotate at different rotational speeds. The rotation axis of the auxiliary or satellite rotor **220** is parallel to and spaced apart from the rotation axis of the upper shaft **110**.

[0088] In a first exemplary embodiment for a position sensor assembly (an inductive sensing type), an auxiliary target **221** having a conductive material such as metal (e.g. aluminum or copper) may be included in or attached to the auxiliary or satellite rotor **220**. The auxiliary target **221** may be an electrically conductive coupler. The auxiliary target **221** may have, for example, but not limited to, a partial circle or polygon shape such as a half circle or a half polygon. The auxiliary target **221** rotates above the auxiliary excitation or transmitter coil set **315** and dissipates the magnetic field generated by the auxiliary excitation or transmitter coil set **315**, thereby creating an imbalance in the auxiliary receiver coil set **314** and consequently generating an output voltage in the auxiliary receiver coil set **314** depending on the angular position of the auxiliary target **221**.

[0089] The auxiliary receiver coil set **314** and the auxiliary excitation or transmitter coil set **315** for sensing the absolute angular rotational position of the upper rotor **210** and/or the lower rotor **230** are included in or disposed on one of both surfaces of the stator **300**, for instance, the upper surface of the stator **300**. For example, the auxiliary receiver coil set **314** and the auxiliary excitation or transmitter coil set **315** may be formed by conductive traces on the upper surface of the stator **300** or electrically conductive pathways on a multi-layer PCB of the stator **300** at a position such that the auxiliary receiver coil set **314** faces the auxiliary target **221**. The auxiliary receiver coil set **314** includes a plurality of oppositely wounded circumferentially adjacent loops which are electrically connected in series with each other. The auxiliary receiver coil set **314** and the auxiliary excitation or transmitter coil set **315** are electrically connected to the controller **500** to output a signal associated with the angular position of the auxiliary target **221** of the auxiliary or satellite rotor **220**. The auxiliary receiver coil set **314** may have any shape such as a substantially sinusoidal or polygonal shape for sensing an absolute angular rotational position. The auxiliary excitation or transmitter coil set **315** may be circularly wound, but the auxiliary excitation or transmitter coil set **315** can have any shape if necessary.

[0090] An upper angular position receiver coil set **313** for sensing the absolute angular rotational position of the upper rotor **210** is included in or disposed on the upper surface of the stator **300**. For example, the upper angular position receiver coil set **313** may be formed by conductive traces on the upper surface of the stator **300** or electrically conductive pathways on a multi-layer PCB of the stator **300** at a position such that the upper angular position receiver coil set **313** faces the upper target **211**. The upper angular position receiver coil set **313** includes a plurality of oppositely wounded circumferentially adjacent loops which are electrically connected in series with each other. The upper angular position receiver coil set **313** is electrically connected to the controller **500** to output a signal associated with an angular position of the upper rotor **210**. For example, the upper angular position receiver coil set **311** may include a sine receiver coil and a cosine receiver coil. The sine receiver coil and the cosine receiver coil included in the upper angular position receiver coil set **311** are surrounded by the excitation or transmitter coil set **312** and/or **322**. The upper angular position receiver coil set **313** may have any shape such as a substantially sinusoidal or polygonal shape for sensing an absolute angular rotational position.

[0091] A lower angular position receiver coil set **323** for sensing the absolute angular rotational position of the lower rotor **230** is included in or disposed on the lower surface of the stator **300**. For example, the lower angular position receiver coil set **323** may be formed by conductive traces on the lower surface of the stator **300** or electrically conductive pathways on a multi-layer PCB of the stator **300** at a position such that the lower angular position receiver coil **323** faces the lower target **231**. The lower angular position receiver coil **323** includes a plurality of oppositely wounded circumferentially adjacent loops which are electrically connected in series with each other. The lower angular position receiver coil **323** is electrically connected to the controller **500** to output a signal associated with an angular position of the lower rotor **230**. For instance, the lower angular position receiver coil set **323** may include a sine receiver coil and a cosine receiver coil. The sine receiver coil and the cosine receiver coil included in the lower angular position receiver coil set **323** are surrounded by the excitation or transmitter coil set **312** and/or **322**. The lower angular position receiver coil set **323** may have any shape such as a substantially sinusoidal or polygonal shape for sensing an absolute angular rotational position.

[0092] In a second exemplary embodiment for a position sensor assembly (a magnet sensing type), a magnetic sensor (e.g. a Hall effect sensor) may be used to detect an absolute angle position of the upper rotor **210** and/or the lower rotor **230**. For example, the auxiliary target **221** may comprise a magnet material such a permanent magnet, and the auxiliary receiver coil set **314** and the auxiliary excitation or transmitter coil **315** may be replaced with a magnetic sensor such as a Hall effect sensor. The magnetic field between the magnet material of the auxiliary target **221** and the magnet sensor **314** can be varied as a function of the angular displacement of the auxiliary target **221** of the auxiliary or satellite rotor **220**.

[0093] Referring to FIG. **4**B, the angular positions of the upper rotor **210** and the auxiliary or satellite rotor **220** are shown over a plurality of rotations, for instance, four rotations. The output signal of the upper angular position receiver coil set **313** associated with the upper target **211** of the upper rotor **210** has a first periodic pattern and the output signal of the auxiliary receiver coil set **314** associated with the auxiliary target **221** of the auxiliary or satellite rotor **220** has a second periodic pattern. The output signal of the upper angular position receiver coil set **313** repeats a first number of times during each revolution of the upper rotor **210**, while the output signal of the auxiliary receiver coil **314** repeats a second number of times during each revolution of the auxiliary or satellite rotor **220**. Therefore, because the output signals of the upper angular position receiver coil **313** and the auxiliary receiver coil **314** overlap only after a specific number of revolutions, the absolute angular rotational position of the upper rotor **210** or the steering wheel **105** can be calculated based on the output signals of the upper angular position receiver coil **313** and the auxiliary receiver coil **314** as programmed by the processor of the controller **500**.

[0094] For instance, by utilizing the Vernier principle through using the mathematical difference or relation between the output signals of the upper angular position receiver coil **313** and the auxiliary receiver coil **314**, the absolute angular rotational position of the upper rotor **210** or the steering wheel **105** can be calculated.

[0095] Likewise, the absolute angular position of the lower rotor **230** may be calculated in a similar way to the calculation of the upper rotor **210** described above.

[0096] FIGS. **1** to **3** illustrate that the auxiliary or satellite rotor **220** is engaged with the upper rotor **210** and is positioned above the stator **300**. However, alternatively or additionally, the auxiliary or satellite rotor **220** can be engaged with the lower rotor **230** and is positioned below the stator **300**.

[0097] In some embodiments of the present disclosure above, the torque sensor assembly and the angle sensor assembly share the same transmitter and the same target (e.g. the same conductive coupler) to save components and reduce possible interference between those two sensor assemblies. However, each of the torque sensor assembly and the angle sensor assembly can have its own transmitter and target.

[0098] FIG. **6** is a block diagram of a controller according to an embodiment of the present disclosure.

[0099] The controller **500** may comprises a first processor **610**, a second processor **620**, an electronic control unit (ECU) **1**, and ECU **2**.

[0100] The first processor **610** comprises the oscillator **400** configured to provide an excitation signal (TX**12**) to a first channel of the excitation or transmitter coil set **312** or **322** which can be inductively associated with the upper target **211** of the upper rotor **210** and the lower target **231** of the lower rotor **230**. A first channel and a second channel of the relative angular displacement receiver coil set **311** and/or **321** for the torque sensor assembly receive electromagnetic signals influenced by the upper target **211** of the upper rotor **210** and the lower target **231** of the lower rotor **230**, and output a first channel relative angular displacement receiver output signal (RXT**1**) and a second channel angular displacement receiver output signal (RXT**2**) to the first processor **610**, respectively. The first processor **610** outputs a first channel torque output signal (T**1**) and a second channel torque output signal (T**2**) to ECU **1** in response to the first channel relative angular displacement receiver output signal (RXT**1**) and the second channel relative angular displacement receiver output signal (RXT**2**). An upper sine angular position receiver coil and an upper cosine angular position receiver coil included in the upper angular position receiver coil set **313** for sensing the absolute angular rotational position of the upper rotor **210** receive electromagnetic signals influenced by the upper target **211** of the upper rotor **210**, and output a first sine angular position receiver output signal (S**1**-RXUR) and a first cosine angular position receiver output signal (C**1**-RXUR) to the first processor **610**, respectively. An auxiliary sine receiver coil and an auxiliary cosine receiver coil included in the auxiliary receiver coil set **314** receive electromagnetic signals influenced by the auxiliary target **221** of the auxiliary or satellite rotor **220**, and output a first auxiliary sine receiver output signal (S**1**-RXS) and a first cosine receiver output signal (C**1**-RXS) to the first processor **610**, respectively. The first processor **610** outputs a first upper target position output signal (P**1**) and a second upper target position output signal (P**2**) to ECU **1** in response to the first sine receiver angular position output signal (S**1**-RXUR), the first cosine angular position receiver output signal (C**1**-RXUR), the first auxiliary sine angular position receiver output signal (S**1**-RXS), and the first cosine angular position receiver output signal (C**1**-RXS).

[0101] The second processor **620** comprises the oscillator **400** configured to provide an excitation signal (TX**34**) to a second channel of the excitation or transmitter coil set **312** or **322** which can be inductively associated with the upper target **211** of the upper rotor **210** and the lower target **231** of the lower rotor **230**. A third channel and a fourth channel of the relative angular displacement receiver coil set **311** and/or **321** for the torque sensor assembly receive electromagnetic signals influenced by the upper target **211** of the upper rotor **210** and the lower target **231** of the lower rotor **230**, and output a third channel relative angular displacement receiver output signal (RXT**3**) and a fourth channel relative angular displacement receiver output signal (RXT**4**) to the second processor **620**, respectively. The second processor **620** outputs a third channel torque output signal (T**3**) and a fourth channel torque output signal (T**4**) to ECU **2** in response to the third channel relative angular displacement receiver output signal (RXT**3**) and the fourth channel relative angular displacement receiver output signal (RXT**4**). An lower sine angular position receiver coil and an lower cosine angular position receiver coil included in the lower angular position receiver coil set **323** for sensing the absolute angular rotational position of the lower rotor **230** receive electromagnetic signals influenced by the lower target **231** of the lower rotor **230**, and output a second sine angular position receiver output signal (S**2**-RXUR) and a second cosine angular position receiver output signal (S**2**-RXUR) to the second processor **620**, respectively. A second auxiliary sine angular position receiver coil and a second auxiliary cosine angular position receiver coil included in the auxiliary receiver coil set **314** receive electromagnetic signals influenced by the auxiliary target **221** of the auxiliary or satellite rotor **220**, and output a second auxiliary sine angular position receiver output signal (S**2**-RXS) and a second cosine angular position receiver output signal (C**2**-RXS) to the second processor **620**, respectively. The second processor **560** outputs a first lower target position output signal (P**3**) and a second lower target position output signal (P**4**) to ECU **2** in response to the second sine angular position receiver output signal (S**2**-RXUR), the second cosine angular position receiver output signal (C**2**-RXUR), the second auxiliary sine angular position receiver output signal (S**2**-RXS), and the second cosine angular position receiver output signal (C**2**-RXS).

[0102] ECU **1** and ECU **2** can calculate the relative angular displacement movement between the upper rotor **210** and the lower rotor **230** using the first channel torque output signal (T**1**), the second channel torque output signal (T**2**), the third channel torque output signal (T**3**), and the fourth channel torque output signal (T**4**) to determine the torque applied to the steering wheel **105** as illustrated in FIG. **4**A, and can calculate the absolute angular positions of the upper rotor **210** and the lower rotor **230** using the first upper target position output signal (P**1**), the second upper target position output signal (P**2**), the first lower target position output signal (P**3**), and the second lower targe position output signal (P**4**) to determine the absolute angular position of the upper rotor **210** and lower rotor **230** as illustrated in FIG. **4**B.

[0103] As discussed above in reference to FIGS. **2**-**6**, the inductive sensor system of embodiments disclosed herein may have, at least, two transmitter coils (e.g., transmitter coil sets **312** and/or **322**) and two receiver coils (e.g., the relative angular displacement receiver coil sets **311** and/or **321**). A placement (e.g., installation position and spacing or the like) of the two transmitter coils may cause mutual coupling between the two transmitter coils, which results in adverse cross talk between the two transmitter coils. Similar adverse cross talk may occur between the two receiver coils and/or any of the other coils (e.g., **312**, **316**, **313**, **322**, **326**, **323**, etc.) shown above in reference to FIGS. **2**-**3**.

[0104] To reduce and/or eliminate such adverse cross talk between these coils (e.g., the transmitter coils, the receiver coils, and/or any of the other coils shown in FIGS. **2**-**3**), the coils must be spaced far apart from one another. Such spacing apart usually requires an in increase in the size (e.g., in surface area, width, etc.) of the PCB (e.g., PCB of the stator **300**), which not only increases the costs of such PCB but also makes installation of such PCB within the limited space inside the vehicle more difficult.

[0105] As a result, embodiments disclosed herein include new coil designs that not only advantageously reduces the adverse cross talk between the coils but also advantageously allows the size of the PCB to remain the same (e.g., not requiring an increase in the size of the PCB for the cross talk to be minimized and/or eliminated).

[0106] In particular, turning now to FIG. **7**, FIG. **7** shows a transparent top-down view of the inductive sensor system according to an embodiment of the present disclosure. As shown in FIG. **7**, the inductive sensor system comprises a receiver coil set **701** (e.g., made up of the relative angular displacement receiver coil sets **311** and/or **321** of FIGS. **2**-**3**), a transmitter coil set **703** having a first transmitter coil **704** and a second transmitter coil **705** (e.g., made up of transmitter coil sets **312** and/or **322** of FIGS. **2**-**3**), and a bias coil set **707** (e.g., made up of the reference coil set **316** and/or **326** of FIGS. **2**-**3**).

[0107] As further shown in FIG. **7**, the transparent top-down view of the inductive sensor system also includes an upper target **709** (e.g., corresponding to upper target **211** of FIG. **2**) and a lower target **711** (e.g., corresponding to lower target **231** of FIG. **3**). In embodiments, the receiver coil set **701** may be positioned radially outside an outer most surface of both the upper target **709** and the lower target **711**. Such a positioning of the receiver coil set **701** advantageously improves a rotational accuracy of the inductive sensor system. Similar to as shown in FIGS. **2**-**3**, the transmitter coil set **703** and the bias coil set **707** are positioned radially inside (e.g., within) the outer most surface of the upper target **709** and of the lower target **711**.

[0108] Although the shape of the targets **709** and **711** in FIG. **7** are different from the shape of the targets **211** and **231** shown in FIGS. **2-3**, this difference in shape will not affect the advantageous effects generated by the shape, configuration, and/or arrangement of the transmitter coil set **703**, receiver coil set **701**, and the bias coil set **707** discussed in reference to FIGS. **8**A-**9**C. For example, the upper target **709** and lower target **711** may both be altered to have the same shape as the targets **211** and **231** shown in FIGS. **2**B-**2**C, or vice versa. Said another way, the targets **211**, **231**, **709**, and **711** may have any type of shape (e.g., any type of shape suitable/used in similar inductive sensor systems and/or as required/defined by a manufacturer/user of the inductive sensor system) without departing from the scope of embodiments disclosed herein.

[0109] Additionally, as also similar to the configuration of embodiments shown in FIGS. **2-3**, the receiver coil set **701**, the transmitter coil set **703**, and the bias coil set **707** may also be disposed (e.g., installed, provided on, etc.) a PCB of the stator **300**. In embodiments, the PCB may be a stationary circuit board positioned between the upper rotor **210** and the lower rotor **230** shown in FIGS. **1-3**.

[0110] Even further, although the receiver coil set **701** and bias coil set **707** are shown in FIG. **7** to be inside of the outer circumferences of the targets **709** and **711**, embodiments disclosed herein are not limited to this configuration. In particular, in some embodiments, both the receiver coil set **701** and the bias coil set **707** may be disposed (i.e., positioned) outside of the outer circumferences of the targets **709** and **711**. In some embodiments, one of the two of the receiver coil set **701** and bias coil set **707** may be disposed within the outer circumferences of the targets **709** and **711** while the other one of the two is disposed outside of the outer circumferences of the targets **709** and **711**.

[0111] Additionally, although the transmitter coil set **703** are shown in FIG. **7** to be under the targets **709** and **711** (e.g., within the outer circumferences of targets **709** and **711**), in some embodiments, the transmitter coil set **703** may be disposed (i.e., positioned) outside of the outer circumferences of the targets **709** and **711**.

[0112] In embodiments, the coils making up the bias coil set **707** may be electrically connected in series with the coils making up receiver coil set **701**. This advantageously cancels any potential direct current (DC) bias generated by any of these coils/coil sets.

[0113] Additional details regarding the shape, configuration, and/or arrangements of each of the coils making up the receiver coil set **701**, the transmitter coil set **703**, and the bias coil set **707** will be discussed below in reference to FIGS. **8**A-**9**D.

[0114] Starting with FIGS. **8**A and **8**B, FIG. **8**A shows a top view of the transmitter coil set **703** according to an embodiment of the present disclosure. FIG. **8**B shows a perspective view of the transmitter coil set **703** according to an embodiment of the present disclosure.

[0115] As seen from the top view shown in FIG. **8**A, each of the first transmitter coil **704** and the second transmitter coil **705** has a first half turn having a first radius R1 and a second half turn having a second radius R2 that is different (e.g., smaller or larger) than the first radius.

[0116] For example, each of the first transmitter coil **704** and the second transmitter coil **705** may have the first radius R**1** in an angle span of 0-180 degrees and the second radius R**2** in an angle span of 180-360 degrees. In embodiments, the value of each of the first radius R**1** and the second radius R**2** may be any number set (e.g., defined by) a manufacturer of the inductive sensor system. In one example, a maximum value (e.g., a maximum radius value for R**1** and R**2**) may be a value ranging between 35-40 mm while a minimum value (e.g., a minimum radius value for R**1** and R**2**) may be a value ranging between 13-17 mm.

[0117] When overlapped over one another (e.g., on the PCB of stator **300**), as shown in FIGS. **8**A and **8**B, the first transmitter coil **704** and the second transmitter coil **705** may be interchangeably spaced to ensure similar electrical properties (e.g., inductance and resistance) for both the first transmitter coil **704** and the second transmitter coil **705**. Such interchangeable spacing also advantageously minimizes and/or eliminates mutual coupling and cross talk between the first transmitter coil **704** and the second transmitter coil **705**.

[0118] As shown in FIGS. **8**A and **8**B, the interchangeably spacing of the first transmitter coil **704** and the second transmitter coil **705** results in a configuration and/or arrangement where (when the first transmitter coil **704** and the second transmitter coil **705** are overlapped on a same plane): (i) the first transmitter coil **704** has the first radius R**1** in an angle span of 0-180 degrees and the second radius R**2** in the angle span of 180-360 degrees; and (ii) the second transmitter coil **705** has the second radius R**2** in the angle span of 0-180 and the first radius R**1** in the angle span of 180-360 degrees.

[0119] In terms of a trace layout perspective on the PCB of stator **300**, assume that the PCB is a two-layer PCB where each layer of the two-layer PCB corresponds to one of two surfaces of the PCB (e.g., the first layer being a first surface and the second layer being a second surface). The first surface of the PCB will have: (i) on a first side of the PCB, the first half turn of the first transmitter coil **704** at the first radius R**1**; and (ii) on a second side opposite to and on the same x-axis (e.g., horizontal) plane as the first side, the second half turn of the second transmitter coil **705** at the second radius R**2**. The second surface of the PCB will then have the opposite (e.g., reversed) layout.

[0120] As a result of such a configuration, each of the first transmitter coil **704** and the second transmitter coil **705** may have (e.g., as shown in FIGS. **8**A-**8**B), a portion where the first radius R**1** transitions into the second radius R**2** and as second portion where the second radius R**2** transitions back into the first radius R**1**. These portions (e.g., transition portions) of the first transmitter coil **704** and the second transmitter coil **705** may cross through the substrate of the PCB (e.g., using one or more vias disposed in the PCB substrate) to allow one part (e.g., one half turn) of the first transmitter coil **704** and the second transmitter coil **705** to be on one surface (e.g., layer) of the PCB and another part (e.g., the other half turn) to be on the other surface (e.g., layer) of the PCB.

[0121] Although the above example discloses a shape and/or configuration where the first transmitter coil **704** and the second transmitter coil **705** are separated into half turns, embodiments disclosed herein are not limited to such a shape and/or configuration. In particular, depending on the number of layers of the PCB, each of the first transmitter coil **704** and the second transmitter coil **705** may be separated into any number of chunks (e.g., quarter turns, third turns, etc.) having any angle span between 0-180 degrees. However, when overlapped over one another on the same plane on the layers of the PCB, the overlapping portions/chunks of the first transmitter coil **704** and the second transmitter coil **705** must have different radius (e.g., if a portion/chunk of the first transmitter coil **704** on the first layer has the first radius R**1**, the corresponding portion/chunk of the second transmitter coil on the second layer that overlaps (e.g., on the z-axis) with the portion/chunk of the first transmitter coil **704** having the first radius R**1** must have a second radius R**2** that is different (e.g., smaller or larger) than the first radius R**1**).

[0122] Additionally, in embodiments, a difference between the first radius R**1** and the second radius R**2** may be, for example, but not limited to the smaller of the two radiuses R**1** and R**2** being at least approximately 0.5 mm to 1.5 mm smaller than the larger of the two radiuses.

[0123] Turning now to FIGS. **9**A-**9**B, FIGS. **9**A-**9**B show a shape and/or configuration of the receiver coil set **701** of FIG. **7** in accordance with one or more embodiments disclosed herein. In embodiments, the receiver coil **701** set may have at least two receiver coils (e.g., a first receiver coil **702**A shown in FIG. **9**A and a second receiver coil **702**B shown in FIG. **9**B. As shown in FIG. **9**C, these two receiver coils may be overlapped (e.g., in a lateral, z-axis manner) across the layers making up the PCB of stator **300**.

[0124] As shown in FIGS. **9**A and **9**B, each of the first receiver coil **702**A and the second receiver coil **702**B may have a coiled structure having M number of full turns (e.g., M number of full rotations of 360 degrees). The M number of full turns may be determined based on a number of layers (N) making up the PCB of stator **300**. For example, the number of full turns (M) may be N divided by 2 (e.g., M=N/2). More specifically, each layer of the PCB may at least have one half turn of at least one of the first receiver coil **702**A or the second receiver coil **702**B.

[0125] Even more specifically, as shown in example of FIG. **9**A where a four (4) layer PCB is used as an example, a first layer (L1) of the PCB has a first half turn from 0-180 degrees at a radius of RL1, the second layer (L2) of the PCB has a second half turn from 180-360 degrees at a radius of RL2, the third layer (L3) of the PCB has a third half turn from 0-180 degrees at a radius RL3, and the fourth layer (L4) of the PCB has a final, fourth half turn from 180-360 at a radius RL4.

[0126] In embodiments, all of RL1 through RL4 of the first receiver coil **702**A may have substantially the same (e.g., accounting more minor deviations due to manufacturing variances, errors, or the like) radius. Alternatively, similar to the configuration of the transmitter coils **704**/**705** shown in FIGS. **8**A-**8**B, each half turn may have different radius where each even layer of the PCB may have a half turn at a first radius R**1** while each odd layer of the PCB may have a half turn at a second radius R**2** that is different (e.g., smaller or larger) than the first radius R**1**.

[0127] In embodiments, as shown in FIG. **9**B, the second receiver coil **702**B may have the exact opposite (e.g., reversed) shape and/or configuration as the first receiver coil **702**B, such that when stacked together within the layers of the PCB, the combined receiver coil set **701** would have a shape and/or configuration as shown in FIG. **9**C where each layer of the PCB would only have opposite half turns of each of the first receiver coil **702**A and the second receiver coil **702**B.

[0128] In embodiments, such an arrangement (e.g., the arrangement shown in FIG. **9**C) and/or the alternative arrange of FIG. **9**C where the receiver coils **702**A/**702**B are interchangeably spaced (e.g., similar to the interchangeably spaced configuration of the transmitter coils **704**/**705**) across the layers of the PCB advantageously makes the receiver coils **702**A/**702**B more robust against any dynamic variations (e.g., dynamic variations in readings of the angular displacement) while also advantageously ensuring that ECU **1** and ECU **2** (in which these coils are provided) will produce (e.g., output) similar (e.g., almost identical with slight variations within a variation threshold set by a manufacturer, or the like) similar gain and/of offset values.

[0129] Although described above as having half turns of each receiver coil **702**A/**702**B, embodiments herein are not limited to such a configuration. In particular, each layer of the PCB can have any amount (X) of a one receiver coil that is less than one full turn and the inverse amount (e.g., 1−X) of the other receiver coil. For example, if one layer of the PCB has a quarter turn of the first receiver coil **702**A, the same layer of the PCB would then have a three-quarter turn of the second receiver coil **702**B.

[0130] In embodiments, the number of full turns of each receiver coil **702**A/**702**B may also be adjusted based on how much of each receiver coil occupies each layer of the PCB. For example, assume that a PCB has four (4) layers and each layer has at least one quarter turn of each receiver coil **702**A/**702**B. In this example, each of the receiver coils **702**A/**702**B would have one full turn that runs from a top-most layer of the PCB (e.g., the first layer) through to the bottom-most layer (e.g., the fourth layer) of the PCB.

[0131] As shown in FIG. **9**D, the first receiver coil **702**A and the second receiver coil **702**B of the receiver coil set **701** may also have a similar configuration as the first transmitter coil **704** and the second transmitter coil **705** of the transmitter coil set **703** where the first receiver coil **702**A and the second receiver coil **702**B are interchangeably spaced (e.g., with varying radiuses) throughout the various layers of the PCB of stator **300**. The bias coils (e.g., the first bias coil and the second bias coil) (not shown individually in the figures) of the bias coil set **707** may also have a similar interchangeably spaced configuration as shown in FIG. **9**D.

[0132] In some embodiments disclosed herein, the bias coil set **707** of FIG. **7** may have any number of bias coils. In one example, the number of bias coils may correspond to the number of receiver coils making up the receiver coil set **701**. More specifically, in this example, if there are two receiver coils (e.g., a first receiver coil **702**A and a second receiver coil **702**B as shown in the above example of FIGS. **9**A-**9**B), there may be exactly two bias coils (e.g., a first bias coil and a second bias coil). In embodiments, the shape and/or configuration of the bias coils (e.g., the layout of the bias coils of the bias coil set on the layers of the PCB of stator **300**) may be identical to that of the receiver coils **702**A and/or **702**B of the receiver coil set **701**.

[0133] Alternatively, in other embodiments disclosed herein the bias coils of the bias coil set **707** may have different shapes and/or configurations than that of the receiver coils of the receiver coil set **701**. In one example, the receiver coils of the receiver coil set **701** may not have the interchangeably spaced configuration with varying radius across the different layers of the PCB (e.g., each full turn of the receiver coils has the same radius) while the bias coils of the bias coil set **707** may have the interchangeably spaced configuration, or vice versa.

[0134] Furthermore, in embodiments, although the receiver coils of the receiver coil set **701** and the bias coils of the bias coil set **707** need not have the interchangeably spaced configuration with varying radius across the different layers of the PCB, the transmitter coils (e.g., **704** and **705**) of the transmitter coil set **703** must have the interchangeably spaced configuration with varying radius across the different layers of the PCB (e.g., the configuration shown in FIGS. **8**A-**8**B and **9**D).

[0135] In embodiments, each bias coil of the bias coil set **707** may have the same number of full turns as each of the receiver coils **702**A/**702**B of the receiver coil set **701**. For example, if the each of the receiver coils **702**A/**702**B has two (2) full turns, each of the bias coils may also have two (2) full turns.

[0136] In embodiments, for a single PCB of the stator **300** and in one example, regardless of the number of layers included in the single PCB, each transmitter coil **704**/**705** of the transmitter coil set **703** for that single PCB may only have a single (e.g., one) full turn. This single full turn of each transmitter coil **704**/**705** may be divided across each of the layers of the PCB in any way. In one example, assume that the PCB has four layers and has the receiver coil set **701** in the same configuration as shown in FIG. **9**C. In this example, a first half turn of the each of the transmitter coils **704**/**705** may be disposed on an upper most layer (e.g., the first layer L1) of the PCB while the second half turn of each of the transmitter coils **704**/**705** may be disposed on a bottom most layer (e.g., the fourth layer L4) of the PCB.

[0137] However, depending on the requirements and/or design preferences of various manufacturers, the single full turn of each of the transmitter coil **704**/**705** may also be spaced out differently between the four (4) layers of the PCB in this example without departing from the scope of embodiments disclosed herein.

[0138] In another example, each of the transmitter coil **704**/**705** may have up to fifteen (15) full turns depending on a number of layers of the PCB of the substrate **300**.

[0139] Any of the above-discussed motor vehicles according to certain exemplary embodiments of the present disclosure may be identical, or substantially similar to, vehicle **800** shown in FIG. **10**. The vehicle **800** may be any passenger or commercial automobile such as a hybrid vehicle, an electric vehicle, or any other type vehicles. FIG. **10** is a schematic view of a vehicle **800** including a steering system and a brake assembly according to an exemplary embodiment of the present disclosure. The vehicle **800** may include a steering system **810** for use in a vehicle. The steering system **810** can allow a driver or operator of the vehicle **800** to control the direction of the vehicle **800** or road wheels **830** of the vehicle **800** through the manipulation of a steering wheel **820**. The steering wheel **820** is operatively coupled to a steering shaft (or steering column) **822**. The steering wheel **820** may be directly or indirectly connected with the steering shaft **822**. For example, the steering wheel **820** may be connected to the steering shaft **822** through a gear, a shaft, a belt and/or any connection means. The steering shaft **822** may be installed in a housing **824** such that the steering shaft **822** is rotatable within the housing **824**.

[0140] The road wheels **830** may be connected to knuckles, which are in turn connected to tie rods. The tie rods are connected to a steering assembly **832**. The steering assembly **832** may include a steering actuator motor **834** and steering rods **836**. The steering rods **836** may be operatively coupled to the steering actuator motor **834** such that the steering actuator motor **834** is adapted to move the steering rods **836**. The movement of the steering rods **836** controls the direction of the road wheels **830** through the knuckles and tie rods.

[0141] One or more sensors **840** (e.g., the inductive sensor system discussed above in reference to FIGS. **1**-**9**C) may be configured to detect position, angular displacement or travel **825** of the steering shaft **822** or steering wheel **820**, as well as detecting the torque of the angular displacement. The sensors **840** provide electric signals to a controller **850** indicative of the angular displacement and torque **825**. The controller **850** sends and/or receives signals to/from the steering actuator motor **834** to actuate the steering actuator motor **834** in response to the angular displacement **825** of the steering wheel **820**.

[0142] In the steer-by-wire steering system, the steering wheel **820** may be mechanically isolated from the road wheels **830**. For example, the steer-by-wire system has no mechanical link connecting the steering wheel **825** from the road wheels **830**. Accordingly, the steer-by wire steering system may comprise a feedback actuator or steering feel actuator **828** comprising an electric motor which is connected to the steering shaft or steering column **822**. The feedback actuator or steering feel actuator **828** provides the driver or operator with the same "road feel" that the driver receives with a direct mechanical link.

[0143] Although the embodiment illustrated in FIG. **10** shows the vehicle **800** having the steer-by-wire steering system, the vehicle **800** may alternatively have a mechanical steering system without departing from embodiments disclosed herein. The mechanical steering system typically includes a mechanical linkage or a mechanical connection between the steering wheel **820** and the road wheels **830**. In the mechanical steering system, the steering actuator motor **834** includes an electric motor to provide power to assist the movement of the road wheels **830** in response to the operation of the driver or a control signal of the controller **850**. Accordingly, the electric motor can be used as the steering actuator motor **834** or can be included in the feedback actuator or steering feel actuator **828**.

[0144] Although the example embodiments have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the application as defined by the appended claims.

[0145] In the present disclosure, relational terms such as first and second, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. Furthermore, depending on the context, words such as "connect" or "coupled to" used in describing a relationship between different elements do not imply that a direct physical connection must be made between these elements. For example, two elements may be connected to each other physically, electronically, logically, or in any other manner, through one or more additional elements. The term "connected" or "coupled" may mean direct or indirect connection unless otherwise specified.

[0146] Plural elements or steps can be provided by a single integrated element or step. Alternatively, a single element or step might be divided into separate plural elements or steps.

[0147] The disclosure of "a" or "one" to describe an element or step is not intended to foreclose additional elements or steps.

[0148] While the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as "first," "second," and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings.

[0149] The terms "comprises," "comprising," "includes," "including," "has," "having," or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a method, article, or apparatus that comprises a list of features is not necessarily limited only to those features but may include other features not expressly listed or inherent to such method, article, or apparatus. Further, unless expressly stated to the contrary, "or" refers to an inclusive-or and not to an exclusive-or. For example, a condition A or B may be satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0150] Although various embodiments of the claimed invention have been described above with a certain degree of particularity, or with reference to one or more individual embodiments, those skilled in the art could make numerous alterations to the disclosed embodiments without departing from the spirit or scope of the claimed invention. The use of the terms "about", "approximately" or "substantially" means that a value of an element has a parameter that is expected to be close to a stated value or position. However, as is well known in the art, there may be minor variations that prevent the values from being exactly as stated. Accordingly, anticipated variances, such as 10% differences, are reasonable variances that a person having ordinary skill in the art would expect and know are acceptable relative to a stated or ideal goal for one or more embodiments of the present disclosure. It is also to be appreciated that the terms "top" and "bottom", "left" and "right", "up" or "down", "first", "second", "before", "after", and other similar terms are used for description and ease of reference purposes only and are not intended to be limiting to any orientation or configuration of any elements or sequences of operations for the various embodiments of the present disclosure.

[0151] Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, and composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure, processes, machines, manufacture, compositions of matter, means, methods or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the embodiments and alternative embodiments. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

What is claimed is:

1. An inductive sensor system comprising:

a circuit board that comprises:

a transmitter coil set comprising at least:

a first transmitter coil comprising a first portion with a first radius and a second portion with a second radius smaller than the first radius; and

a second transmitter coil comprising a first portion with a third radius identical to the first radius and a second portion with a fourth radius identical to the second radius,

wherein the first transmitter coil and the second transmitter coil are circularly wound in a manner where the first radius and the fourth radius are on a first side of the circuit board while the second radius and the third radius are on a second side of the circuit board; and

a receiver coil set comprising at least a first receiver coil and a second receiver coil, wherein the first receiver coil and the second receiver coil are circularly wound.

2. The inductive sensor system of claim **1**, wherein

the receiver coil set surrounds the transmitter coil set,

the circuit board comprises at least a first surface and a second surface opposite to the first surface,

the first surface on the first side of the circuit board comprises the first portion of the first transmitter coil,

the first surface on the second side of the circuit board comprises the first portion of the second transmitter coil,

the second surface on the first side of the circuit board comprises the second portion of the second transmitter coil, and

the second surface on the second side of the circuit board comprises the second portion of the first transmitter coil.

3. The inductive sensor system of claim **2**, wherein

the first transmitter coil further comprises a third portion where the first radius transitions into the second radius and a fourth portion where the second radius transitions back into the first radius,

the second transmitter coil further comprises a third portion where the third radius transitions into the fourth radius and a fourth portion where the fourth radius transitions back into the third radius, and

the third portion of the first transmitter coil overlaps with the third portion of the second transmitter coil and the fourth portion of the first transmitter coil overlaps with the fourth portion of the second transmitter coil within a body of the circuit board that separates the first surface from the second surface.

4. The inductive sensor system of claim **3**, further comprising:

an upper rotor comprising an upper target having a first metallic pattern; and

a lower rotor comprising a lower target having a second metallic pattern, wherein the circuit board is a stationary circuit board that is positioned between the upper rotor and the lower rotor, the transmitter coil set is configured to generate an electromagnetic field, and the receiver coil set is configured to sense relative angular displacement movement between the upper rotor and the lower rotor.

5. The inductive sensor system of claim **4**, wherein

the circularly wound receiver coil set is positioned radially outside an outer most surface of the first metallic pattern and of the second metallic pattern, and

the circularly wound transmitter coil set is positioned radially within the outer most surface of the first metallic pattern and of the second metallic pattern.

6. The inductive sensor system of claim **5**, wherein the first metallic pattern of the upper target and/or the second metallic pattern of the lower target have a plurality of circumferentially adjacent lobes.

7. The inductive sensor system of claim **4**, wherein the circuit board further comprises:

a bias coil set comprising at least a first bias coil and a second bias coil, wherein the first bias coil and the second bias coil are circularly wound and the bias coil set is surrounded by the transmitter coil set.

8. The inductive sensor system of claim **7**, wherein

the circuit board comprises N number of layers,

each of the first receiver coil and the second receiver coil comprises a first coiled structure comprising N divided by 2 number of full turns, and

each of the layers of the circuit board comprises at least a half turn of each of the first receiver coil and the second receiver coil.

9. The inductive sensor system of claim **8**, wherein the half turn of the first receiver coil and of the second receiver coil on a same layer of the layers comprise a same radius.

10. The inductive sensor system of claim **8**, wherein the half turn of the first receiver coil and of the second receiver coil on a same layer of the layers comprise different radiuses.

11. The inductive sensor system of claim **8**, wherein

each of the first bias coil and the second bias coil comprises a second coiled structure comprising N divided by 2 number of full turns, and

each of the layers of the circuit board comprises at least a half turn of each of the first bias coil and the second bias coil.

12. The inductive sensor system of claim **11**, wherein the half turn of each of the first bias coil and the second bias coil on a same layer of the layers comprise a same radius.

13. The inductive sensor system of claim **11**, wherein the half turn of each of the first bias coil and the second bias coil on a same layer of the layers comprise different radiuses.

14. The inductive sensor system of claim **11**, wherein a total first number of the full turns of the receiver coil set is identical to a second total number of the full turns of the bias coil set.

15. The inductive sensor system of claim **8**, wherein the bias coil set is disposed in series with the receiver coil set and is configured to cancel DC bias generated by the receiver coil set.

16. A motor vehicle comprising:

one or more road wheels;

a steering wheel coupled to at least one of the one or more road wheels; and

an inductive sensor system coupled to the steering wheel, the inductive sensor system comprising:

a circuit board that comprises:

a transmitter coil set comprising at least:

a first transmitter coil comprising a first portion with a first radius and a second portion with a second radius smaller than the first radius; and

a second transmitter coil comprising a first portion with a third radius identical to the first radius and a second portion with a fourth radius identical to the second radius,

wherein the first transmitter coil and the second transmitter coil are circularly wound in a manner where the first radius and the fourth radius are on a first side of the circuit board while the second radius and the third radius are on a second side of the circuit board; and

a receiver coil set comprising at least a first receiver coil and a second receiver coil, wherein the first receiver coil and the second receiver coil are circularly wound.

17. The motor vehicle of claim **16**, wherein

the receiver coil set surrounds the transmitter coil set,

the circuit board comprises at least a first surface and a second surface opposite to the first surface,

the first surface on the first side of the circuit board comprises the first portion of the first transmitter coil,

the first surface on the second side of the circuit board comprises the first portion of the second transmitter coil,

the second surface on the first side of the circuit board comprises the second portion of the second transmitter coil, and

the second surface on the second side of the circuit board comprises the second portion of the first transmitter coil.

18. The motor vehicle of claim **17**, wherein

the first transmitter coil further comprises a third portion where the first radius transitions into the second radius and a fourth portion where the second radius transitions back into the first radius,

the second transmitter coil further comprises a third portion where the third radius transitions into the fourth radius and a fourth portion where the fourth radius transitions back into the third radius,

wherein the third portion of the first transmitter coil overlaps with the third portion of the second transmitter coil and the fourth portion of the first transmitter coil overlaps with the fourth portion of the second transmitter cocmil within a body of the circuit board that separates the first surface from the second surface.

19. The motor vehicle of claim **16**, wherein the inductive sensor system further comprises:

an upper rotor comprising an upper target having a first metallic pattern; and

a lower rotor comprising a lower target having a second metallic pattern,

wherein the circuit board is a stationary circuit board that is positioned between the upper rotor and the lower rotor, the transmitter coil set is configured to generate an electromagnetic field, and the receiver coil set is configured to sense relative angular displacement movement between the upper rotor and the lower rotor.

20. The motor vehicle of claim **19**, wherein

the upper rotor is comprised in or coupled to an upper shaft coupled to the steering wheel,

the lower rotor is comprised in or coupled to a lower shaft, and

a torsion bar is coupled between the upper shaft and the lower shaft.

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